

Solutions

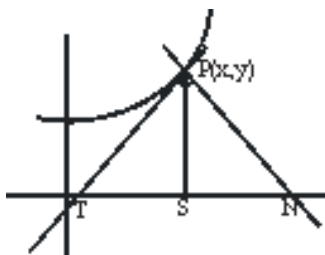
Maths

Q1. If area of triangle formed by tangent (with positive slope) and normal to a curve at any point in first quadrant with x-axis is cube of its ordinate, then differential equation of such family is :

- (A) $\left(\frac{dy}{dx}\right)^2 = 2y\frac{dy}{dx}$ (B) $\left(\frac{dy}{dx}\right)^2 - 2y\frac{dy}{dx} + 1 = 0$ (C) $2y\left(\frac{dy}{dx}\right)^2 + 1 = 0$
 (D) $\left(\frac{dy}{dx}\right)^2 + 2y\frac{dy}{dx} + 1 = 0$

Ans. B

Sol.



$$\begin{aligned}\text{Area} &= \frac{1}{2}(ST + SN) \times PS \\ y^3 &= \frac{1}{2}\left(y y' + \frac{y}{y'}\right) \cdot y \\ \Rightarrow 2y &= y' + \frac{1}{y'} \\ \Rightarrow \left(\frac{dy}{dx}\right)^2 - 2y\left(\frac{dy}{dx}\right) + 1 &= 0\end{aligned}$$

Q2. Let $f(x)$ be a function satisfying $f'(x) = \frac{x^3 - f(x)}{x}$ with $f(2) = 2$, then the value of $\int_1^2 e^{x^2} \cdot f(x) dx$ is :

- (A) $\frac{5e^2}{8}$ (B) $\frac{e^2}{8}$ (C) $\frac{3e^4}{8}$ (D) $\frac{5}{8}3e^4$

Ans. C

$$\begin{aligned}\text{Sol. } x f'(x) + f(x) &= x^3 \\ \Rightarrow f(x) \cdot e^{\int \frac{1}{x} dx} &= \int x^2 \cdot e^{\int \frac{1}{x} dx} dx \\ \Rightarrow x f(x) &= \int x^3 dx \\ \Rightarrow x f(x) &= \frac{x^4}{4} + C\end{aligned}$$

given $f(2) = 2 \Rightarrow c = 0$

$$\therefore f(x) = \frac{x^3}{4}$$

$$I = \int_1^2 e^{x^2} f(x) dx = \int_1^2 e^{x^2} \cdot \frac{x^2 \cdot x}{4} dx = \frac{3e^4}{8}$$

Q3. Solution of differential equation $\frac{dy}{dx} = \frac{1}{xy(x^2 \sin y^2 + 1)}$ is (where C is integral constant) :

$$(A) e^{y^2} \left(\frac{1}{x^2} - \frac{\cos y^2}{2} + \frac{\sin y^2}{2} \right) = C \quad (B) e^{y^2} \left(\frac{1}{x^2} + \frac{\cos y^2}{2} - \frac{\sin y^2}{2} \right) = C$$

$$(C) e^{y^2} \left(\frac{1}{x^2} - \frac{\cos y^2}{2} + \frac{\sin^2 y}{2} \right) = C \quad (D) e^{y^2} \left(\frac{1}{x^2} - \frac{\cos y}{2} + \frac{\sin y}{2} \right) = C$$

Ans. A

Sol. Let $y^2 = t \Rightarrow 2y \frac{dy}{dx} = \frac{dt}{dx}$

$$\frac{1}{2} \frac{dt}{dx} = \frac{1}{x(x^2 \sin t + 1)}$$

$$\frac{dx}{dt} = \frac{x^3 \sin t}{2} + \frac{x}{2}$$

$$x^{-3} \frac{dx}{dt} = \frac{\sin t}{2} + \frac{x^{-2}}{2}$$

Let $x^{-2} = z \Rightarrow x^{-3} \frac{dx}{dt} = -\frac{dz}{2dt}$

$$-\frac{1}{2} \frac{dz}{dt} = \frac{\sin t}{2} + \frac{z}{2}$$

$$\frac{dz}{dt} + z = -\sin t$$

$$ze^t = -\int e^t \sin t dt$$

$$ze^t = -\frac{(-e^t \cos t + e^t \sin t)}{2} + C$$

$$e^{y^2} \left(\frac{1}{x^2} - \frac{\cos y^2}{2} + \frac{\sin y^2}{2} \right) = C$$

Q4. The solution of the differential equation $2x^2 y \frac{dy}{dx} = \tan(x^2 y^2) - 2xy^2$ given $y(1) = \sqrt{\frac{\pi}{2}}$ is :

$$(A) \sin(x^2 y^2) - 1 = 0 \quad (B) \cos\left(\frac{\pi}{2} + x^2 y^2\right) + x = 0 \quad (C) \sin(x^2 y^2) = e^{x-1}$$

$$(D) \sin(x^2 y^2) = e^{2(x-1)}$$

Ans. C

Sol. Let $x^2 y^2 = t$

$$y(1) = \sqrt{\frac{\pi}{2}} \Rightarrow \text{at } x = 1, t = \frac{\pi}{2}$$

$$2xy^2 + 2x^2 y \frac{dy}{dx} = \frac{dt}{dx}$$

$$\frac{dt}{dx} = \tan t$$

$$\Rightarrow \int \cot(t) dt = \int dx$$

$$\begin{aligned}
&\Rightarrow \ln |\sin t| = x + \ln c \\
&\Rightarrow \sin t \\
&= ce^x \\
&\Rightarrow c \\
&= e^{-1} \\
&\Rightarrow \sin(x^2 y^2) \\
&= e^{x-1}
\end{aligned}$$

Q5. General solution of differential equation of $f(x) \frac{dy}{dx} = f^2(x) + f(x)y + f'(x)y$ is (c being arbitrary constant) :

(A) $y = f(x) + ce^x$ (B) $y = -f(x) + ce^x$ (C) $y = -f(x) + ce^x f(x)$ (D) $y = cf(x) + e^x$

Ans. C

Sol. $\frac{dy}{dx} - \left(1 + \frac{f'(x)}{f(x)}\right)y = f(x)$
 $= e^{\int -\left(1 + \frac{f'(x)}{f(x)}\right)dx}$
I.F. $= e^{-x - \ln f(x)}$
 $= \frac{e^{-x}}{f(x)}$
 $\frac{ye^{-x}}{f(x)} = \int e^{-x} dx + C$
 $\Rightarrow \frac{ye^{-x}}{f(x)} = -e^{-x} + C$

Q6. Let $f(x)$ be differentiable function on the interval $(0, \infty)$ such that $f(1) = 1$ and $\lim_{t \rightarrow x} \left(\frac{t^3 f(x) - x^3 f(t)}{t^2 - x^2} \right) = \frac{1}{2}$. If $x > 0$, then $f(x)$ is :

(A) $\frac{1}{4x} + \frac{3x^2}{4}$ (B) $\frac{3}{4x} + \frac{x^3}{4}$ (C) $\frac{1}{4x} + \frac{3x}{4}$ (D) $\frac{1}{4x^3} + \frac{3x}{4}$

Ans. C

Sol. $\lim_{t \rightarrow x} \frac{t^3 f(x) - x^3 f(t)}{t^2 - x^2} = \frac{1}{2}$
 $3xf(x) - x^2 f'(x) = 1$
 $\frac{dy}{dx} - \frac{3}{x}y = -\frac{1}{x^2}$
 $y = \frac{1}{4x} + \frac{3}{4}x^2 \quad (\because f(1) = 1)$

Q7. Solution of the differential equation $\frac{dy}{dx} = \frac{5x^3 - xy^2 - 2x}{3x^2y - y^3}$, is:

(A) $\ln |y^4 - 4x^2y^2 + 5x^4 + 2x^2 - 2y^2 + 2| - \tan^{-1} \left(\frac{y^2 - 2x^2 - 1}{x^2 - 1} \right) = c$
(B) $\ln |y^4 - 4x^2y^2 + 5x^4 + 2x^2 - 2y^2 + 2| + \tan^{-1} \left(\frac{y^2 - 2x^2 - 1}{x^2 - 1} \right) = c$
(C) $\ln |y^4 - 4x^2y^2 + 5x^4 + 2x^2 - 2y^2 + 2| + 2 \tan^{-1} \left(\frac{y^2 - 2x^2 - 1}{x^2 - 1} \right) = c$

$$(D) \ln |y^4 - 4x^2y^2 + 5x^4 + 2x^2 - 2y^2 + 2| - 2 \tan^{-1} \left(\frac{y^2 - 2x^2 - 1}{x^2 - 1} \right) = c$$

Ans. D

$$\text{Sol. } \frac{dy}{dx} = \frac{x}{y} \frac{5(x^2 - 1) - (y^2 - 3)}{3(x^2 - 1) - (y^2 - 3)}$$

On putting $(y^2 - 3) = v(x^2 - 1)$, we get

$$\begin{aligned} 2xv + (x^2 - 1) \frac{dv}{dx} &= 2y \frac{dy}{dx} = 2x \left(\frac{5 - v}{3 - v} \right) \\ \Rightarrow \frac{(x^2 - 1)}{2x} \frac{dv}{dx} &= \frac{v^2 - 4v + 5}{3 - v} \Rightarrow \int \frac{v - 3}{v^2 - 4v + 5} dv + \int \frac{2x}{x^2 - 1} dx = 0 \\ \Rightarrow \frac{1}{2} \ell n |v^2 - 4v + 5| - \tan^{-1}(v - 2) + \ell n |x^2 - 1| &= c \end{aligned}$$

Q8. A curve $y = f(x)$ passes through the point P(1,1). The normal to the curve at P is $a(y - 1) + (x - 1) = 0$. If the slope of the tangent at any point on the curve is proportional to the ordinate of the point, then the equation of tangent to the curve at point $Q(0, e^{-a})$ is :

$$(A) y = \pm e^{-a}(ax + 1) \quad (B) y = -e^{-a}(ax + 1) \quad (C) y = e^{-a}(ax + 1) \quad (D) y = e^{-a}(x + a)$$

Ans. C

$$\text{Sol. } \text{Slope of tangent at } Q(0, e^{-a}) \text{ is } \left(\frac{dy}{dx} \right)_Q = ae^{-a}$$

$$\Rightarrow y - e^{-a} = ae^{-a}(x)$$

$$\Rightarrow y = e^{-a}(ax + 1)$$

Q9. $\int e^x \left(\frac{1}{\ln x} - \frac{1}{(\ln x)(\ln x^x)} \right) dx$ is equal to : (where C is constant of integration)

$$(A) \frac{e^x}{\ln x} + C \quad (B) e^x \ln x + C \quad (C) 2e^x \cdot \log_e x + C \quad (D) \frac{e^x}{(\ln x)^2} + C$$

Ans. A

$$\begin{aligned} \text{Sol. } & \int e^x \left(\frac{1}{\ln x} - \frac{1}{x(\ln x)^2} \right) dx \\ &= \int \frac{e^x}{\ln x} dx - \int \frac{e^x}{x(\ln x)^2} dx \\ &= \left[\frac{1}{\ln x} e^x - \int \frac{-1}{(\ln x)^2 x} e^x dx \right] - \int \frac{e^x}{x(\ln x)^2} dx + C \\ & \text{(Applying integration by parts)} \\ &= \frac{e^x}{\ln x} + C \end{aligned}$$

Q10. Let $\int f(\ln x) dx = xe^x + c$, then $\int \frac{f(x) \cdot e^x}{(1 + e^x)} dx$ is :

$$(A) e^x + c \quad (B) \ln x \cdot e^x + c \quad (C) e^x \cdot e^x + c \quad (D) xe^x + c$$

Ans. A

Sol. Differentiate both the sides

$$f(\ln x) = e^x + xe^x$$

$$\text{Let } \ln x = t \Rightarrow x = e^t$$

$$\int \frac{f(x)e^x}{(1+e^x)} dx = \int e^x \cdot e^x dx$$

$$\text{Let } e^x = t \Rightarrow e^x dx = dt$$

$$\int e^t dt = e^t + c = e^{e^x} + c$$

Q11. $\int x e^{\tan x} (2 + x \sec^2 x) dx$ is equal to :

- (A) $2e^{\tan x} + C$ (B) $x^2 e^{\tan x} + C$ (C) $\tan x + x e^{\tan x} + C$ (D) $x^2 + x e^{\tan x} + C$

Ans. B

Sol. Let $I = \int ((x e^{\tan x} + x e^{\tan x} (1 + \sec^2 x)))$
 $= \int (f(x) + x f'(x)) dx = x f(x) + c$
 $= x^2 e^{\tan x} + C$

Q12. $\int \frac{dx}{\sqrt{1 - \tan^2 x}} = \frac{1}{\lambda} \sin^{-1}(\lambda \sin x) + C$, then $\lambda =$:

- (A) $\sqrt{2}$ (B) $\sqrt{3}$ (C) 2 (D) $\sqrt{5}$

Ans. A

Sol. $\int \frac{dx}{\sqrt{1 - \tan^2 x}}$
 $\int \frac{\cos x dx}{\sqrt{1 - 2 \sin^2 x}} = \frac{1}{\sqrt{2}} \sin^{-1}(\sqrt{2} \sin x) + c$

Q13. $\int \frac{(2012)^x}{\sqrt{1 - (2012)^{2x}}} (2012)^{\sin^{-1}(2012)^x} dx =$:

(where C denotes arbitrary constant)

- (A) $(\log_{2012} e)(2012)^{\sin^{-1}(2012)^x} + C$ (B) $(\log_{2012} e)^2 (2012)^{x + \sin^{-1}(2012)^x} + C$
(C) $(\log_{2012} e)^2 (2012)^{\sin^{-1}(2012)^x} + C$ (D) $\frac{(2012)^{\sin^{-1}(2012)^x}}{(\log_{2012} e)^2} + C$

Ans. C

Sol. $\int \frac{(2012)^x}{\sqrt{1 - (2012)^{2x}}} (2012)^{\sin^{-1}(2012)^x} . dx$
 $\Rightarrow \frac{1}{\ln 2012} \int (2012)^t . dt$
Let $\sin^{-1}(2012)^x = t$
 $= \frac{(2012)^{\sin^{-1}(2012)^x}}{\ln^2(2012)} + C$

Q14. If y is implicit differentiable function of x such that $y(x+y)^2 = x$ then $\int \frac{dx}{x+3y} =$:

- (A) $\ln((x+y)^2 + 1) + c$ (B) $\frac{1}{2} \ln((x-y)^2 + 1) + c$ (C) $\ln((x-y)^2 + 1) + c$
(D) $\frac{1}{2} \ln((x+y)^2 + 1) + c$

Ans. D

Sol. $y(x+y)^2 = x$
 $\frac{dy}{dx} = \frac{1-2y(x+y)}{(x+y)(x+3y)}$
 $\Rightarrow 1 + \frac{dy}{dx} = \frac{(x+y)^2 + 1}{(x+y)(x+3y)}$
 $\frac{(x+y)(dx+dy)}{(x+y)^2 + 1} = \frac{dx}{x+3y} \quad \therefore \int \frac{dx}{x+3y}$
 $= \frac{1}{2} \ln((x+y)^2 + 1) + c$

Q15. The value of D. I. $\int_{-2010}^{2010} x^{2010} \cot^{-1}(2010x) dx$ is equal to :

(A) $\frac{\pi}{2011}(2011)^{2010}$ (B) $\frac{\pi}{2010}(2010)^{2011}$ (C) $\frac{\pi}{2011}(2010)^{2011}$ (D) $\frac{\pi}{2010}(2011)^{2010}$

Ans. C

Sol. $I = \int_{-2010}^{2010} x^{2010} \cos^{-1}(2010x) dx$
 $I = \int_{-2010}^{2010} x^{2010} \cos^{-1}(-2010x) dx$
 $2I = \int_{-2010}^{2010} x^{2010} (\pi) dx$

Q16. Let $I = \int_0^{\frac{\pi}{4}} \cos^8 \theta d\theta$, then :

(A) $\frac{\pi}{64} < I < \frac{\pi}{4}$ (B) $I < \frac{\pi}{64}$ (C) $I > \frac{\pi}{4}$ (D) $I = \frac{\pi}{64}$

Ans. A

Sol. $0 < \theta < \frac{\pi}{4} \Rightarrow \frac{1}{16} < \cos^8 \theta < 1$
 $\therefore \int_0^{\frac{\pi}{4}} \cos^8 \theta d\theta \in \left(\frac{\pi}{64}, \frac{\pi}{4} \right)$

Q17. Let $f(x) = \int_{x^2}^{x^3} \frac{dt}{\ln t}$ for $x > 1$ and $g(x) = \int_1^x (2t^2 - \ln t) f(t) dt$ ($x > 1$), then :

- (A) g is increasing on $(1, \infty)$ (B) g is decreasing on $(1, \infty)$
(C) g is increasing on $(1, 2)$ and decreasing on $(2, \infty)$
(D) g is decreasing on $(1, 2)$ and increasing on $(2, \infty)$

Ans. A

Sol. $g'(x) = (2x^2 - \ln x) f(x)$
 $f'(x) = \frac{1}{\ln x^3} 3x^2 - \frac{1}{\ln x^2} 2x$
 $f'(x) = \frac{x^2 - x}{\ln x}$
 $f'(x) = \frac{x(x-1)}{\ln x} > 0 \quad \forall x > 1 ; f(x) > f(1) \Rightarrow f(x) > 0 \quad \forall x > 1$

for $g(x)$ is increasing

$$g'(x) > 0 \Rightarrow 2x^2 - \ln x > 0 \text{ as } (f(x) > 0)$$

$$\text{let } H(x) = 2x^2 - \ln x$$

$$H'(x) = 4x - 1/x$$

$$= \frac{4x^2 - 1}{x} > 0 \text{ when } x > 1$$

$$H(x) > H(1) \Rightarrow H(x) > 2$$

$$\therefore g'(x) > 0 \quad \forall x \in (1, \infty)$$

$$\therefore g(x) \text{ is increasing on } (1, \infty)$$

Q18. Maximum value of the function $f(x) = \pi^2 \int_0^1 t \sin(x + \pi t) dt$ over all real number x :

(A) $\sqrt{\pi^2 + 1}$ (B) $\sqrt{\pi^2 + 2}$ (C) $\sqrt{\pi^2 + 2}$ (D) $\sqrt{\pi^2 + 4}$

Ans. D

Sol.
$$f(x) = \pi^2 \left(\left[\frac{-t \cos(x + \pi t)}{\pi} \right]_0^1 + \int_0^1 \frac{1 \cdot \cos(x + \pi t)}{\pi} dt \right)$$

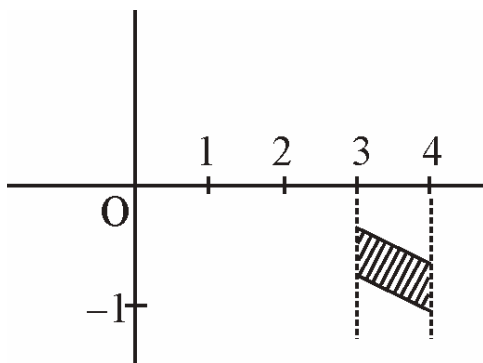
$$= \pi \cos x - 2 \sin x$$

Q19. The area enclosed by the curve $[x + 3y] = [x - 2]$ where $x \in [3, 4]$ is where $[.]$ denote greatest integer function :

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) 1

Ans. B

Sol.



$$1 \leq x + 3y < 2$$

$$\text{Area of shaded region} = \frac{1}{3}$$

Q20. Consider a function $f(x) = \int_0^x [\cos t] dt$, $x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$, then area bounded by curve $y = f(x)$, x-axis and lines $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$ is (where $[.]$ denotes greatest integer function) :

(A) π (B) π^2 (C) $\frac{\pi^2}{2} - 1$ (D) $\frac{\pi^2}{2}$

Ans. D

Sol.

$$f(x) = \int_0^{\frac{\pi}{2}} [\cos t] dt + \int_{\frac{\pi}{2}}^0 [\cos t] dt, x \in \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$$

$$\Rightarrow f(x) = 0 + \int_{\frac{\pi}{2}}^x -1 dt = \frac{\pi}{2} - x$$

$$\Rightarrow \text{Required area} = \frac{1}{2} \times AB \times BC = \frac{\pi^2}{2}$$

Q21. Given $\int_0^y e^t dt + \int_0^x f(t) dt = g(x)$, then $\frac{dy}{dx}$ is equal to (where $f(x)$ is continuous and $g(x)$ is a differentiable function)

- (A) $\frac{g'(x) - f(x)}{1 + g(x) - \int_0^x f(t) dt}$ (B) $\frac{g'(x) - f(x)}{g(x) - \int_0^x f(t) dt}$ (C) $e^y(g'(x) - f(x))$
 (D) $\frac{(g'(x) - f(x)) \cdot e^y}{g(x) - \int_0^x f(t) dt}$

Ans. A

Sol. $\int_0^y e^t dt + \int_0^x f(t) dt = g(x) \dots(1)$

diff. w.r.t. x

$$e^y \frac{dy}{dx} + f(x) = g'(x)$$

$$\Rightarrow \frac{dy}{dx} = \frac{g'(x) - f(x)}{e^y} \dots(2)$$

Again (1)

$$\Rightarrow e^y - 1 = g(x) - \int_0^x f(t) dt$$

$$\Rightarrow e^y = 1 + g(x) - \int_0^x f(t) dt$$

$$\therefore (2) \Rightarrow \frac{dy}{dx} = \frac{g'(x) - f(x)}{1 + g(x) - \int_0^x f(t) dt}$$

Q22. Area bounded by $x^2y^2 + y^4 - x^2 - 5y^2 + 4 = 0$ is equal to :

- (A) $\frac{4\pi}{3} + \sqrt{2}$ (B) $\frac{4\pi}{3} - \sqrt{2}$ (C) $\frac{4\pi}{3} + 2\sqrt{3}$ (D) none of these

Ans. C

Sol. $x^2y^2 + y^4 - x^2 - 5y^2 + 4 = 0$
 $(x^2 + y^2 - 4)(y^2 - 1) = 0$

Q23. A function $f(x)$ satisfy $f(x) = f\left(\frac{4}{x}\right)$ for $x > 0$ if $\int_1^2 \frac{f(x)}{x} dx = 3$, then value of $\int_1^4 \frac{f(x)}{x} dx$ is :

- (A) 0 (B) 6 (C) -3 (D) 3

Ans. B

Sol.
$$\int_1^4 \frac{f(x)}{x} dx$$
$$\int_1^2 \frac{f(x)}{x} dx + \int_2^4 \frac{f(x)}{x} dx$$
$$\int_1^2 \frac{f(x)}{x} dx + \int_1^2 \frac{f\left(\frac{4}{x}\right)}{x} dx = 6$$

Q24. The value of the constant $a > 0$ such that $\int_0^a [\tan^{-1} \sqrt{x}] dx = \int_0^a [\cot^{-1} \sqrt{x}] dx$, where $[\cdot]$ denotes the greatest integer function is :

(A) $\frac{2(3 + \cos 4)}{(1 - \cos 4)}$ (B) $\frac{2(3 - \cos 4)}{(1 + \cos 4)}$ (C) $\frac{2(3 - \cos 4)}{(1 - \cos 4)}$ (D) $\frac{2(3 + \cos 4)}{(1 + \cos 4)}$

Ans. A

Sol. The given integrals can be equal only if $a > \tan^2 1$

$$\begin{aligned} \therefore \int_0^a [\tan^{-1} \sqrt{x}] dx &= \int_0^{\tan^2 1} [\tan^{-1} \sqrt{x}] dx + \int_{\tan^2 1}^a [\tan^{-1} \sqrt{x}] dx \\ &= 0 + 1 \int_{\tan^2 1}^a dx \\ &= a - \tan^2 1 \\ \int_0^a [\cot^{-1} \sqrt{x}] dx &= \int_0^{\tan^2 1} [\cot^{-1} \sqrt{x}] dx + 0 \\ &= 1(\cot^2 1 - 0) \\ \therefore a - \tan^2 1 &= \cot^2 1 \\ \Rightarrow a &= \tan^2 1 + \cot^2 1 \\ &= \frac{\sin^2 1}{\cos^2 1} + \frac{\cos^2 1}{\sin^2 1} \\ &= \frac{1 - \cos 2}{1 + \cos 2} + \frac{1 + \cos 2}{1 - \cos 2} \\ &= \frac{(1 + \cos 2)^2 + (1 - \cos 2)^2}{(1 + \cos 2)(1 - \cos 2)} \\ &= \frac{2(1 + \cos^2 2)}{\sin^2 2} \\ &= \frac{2(2 + 2\cos^2 2)}{2\sin^2 2} \\ &= \frac{2(3 + \cos 4)}{1 - \cos 4} \end{aligned}$$

Q25. If $f(1) = e^2$ and $\frac{d}{dx}(f(x)) > 3x^2 f(x) \forall x \geq 1$ then $f(x)$ cannot take values :

(A) e (B) $\frac{9}{2}$ (C) 8 (D) 10

Ans. A, B

Sol.
$$f'(x) - 3x^2 f(x) > 0 \forall x \geq 1$$
$$\Rightarrow e^{-x^3} f'(x) - 3x^2 e^{-x^3} f(x) > 0$$
$$\Rightarrow \frac{d}{dx}(e^{-x^3} f(x)) > 0 \forall x \geq 1$$

$$\begin{aligned}
&\Rightarrow e^{-x^3} f(x) \\
&\geq e^{-1} f(1) \\
&\Rightarrow e^{-x^3} f(x) \\
&\geq e^2 \\
&\Rightarrow f(x) \geq e^{x^3+1} \forall x \geq 1 \\
&f(x) \geq e^2
\end{aligned}$$

Q26. $\int \frac{\sec^2 x}{\sqrt{a \sec^2 x - b \tan^2 x}} dx$ is : (where c is integration constant)

- (A) $\frac{1}{\sqrt{b-a}} \sin^{-1} \left(\tan x \sqrt{\frac{b-a}{a}} \right) + c$ if $b > a > 0$
- (B) $\frac{1}{\sqrt{b-a}} \log_e \left(\tan x \sqrt{b-a} + \sqrt{a \sec^2 x - b \tan^2 x} \right) + c$ if $b > a > 0$
- (C) $\frac{1}{\sqrt{a-b}} \sin^{-1} \left(\tan x \sqrt{\frac{a-b}{a}} \right) + c$ if $a > b > 0$
- (D) $\frac{1}{\sqrt{a-b}} \log_e \left(\tan x \sqrt{a-b} + \sqrt{a \sec^2 x - b \tan^2 x} \right) + c$ if $a > b > 0$

Ans. A, D

Sol.
$$\begin{aligned}
&\int \frac{\sec^2 x}{\sqrt{a \sec^2 x - b \tan^2 x}} dx \\
&= \int \frac{\sec^2 x}{\sqrt{a(1 + \tan^2 x) - b \tan^2 x}} dx
\end{aligned}$$

put $\tan x = t$

$$\sec^2 x dx = dt$$

$$= \int \frac{dt}{\sqrt{a + t^2(a-b)}}$$

we know

$$\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1} \left(\frac{x}{a} \right) + c$$

$$\int \frac{dx}{\sqrt{a^2 + x^2}} = \log_e \left(x + \sqrt{x^2 + a^2} \right) + c$$

If $a > b > 0$

$$\begin{aligned}
&\int \frac{dt}{\sqrt{(\sqrt{a})^2 + (t\sqrt{a-b})^2}} \\
&= \frac{1}{\sqrt{a-b}} \log_e \left(t\sqrt{a-b} + \sqrt{a + t^2(a-b)} \right) \\
&= \frac{1}{\sqrt{a-b}} \log_e \left(\tan x \sqrt{a-b} + \sqrt{a + \sec^2 x - b \tan^2 x} \right) + c
\end{aligned}$$

If $b > a > 0$

$$\begin{aligned}
&\int \frac{dt}{\sqrt{(\sqrt{a})^2 - (t\sqrt{b-a})^2}} = \frac{1}{\sqrt{b-a}} \sin^{-1} \left(\frac{t\sqrt{b-a}}{\sqrt{a}} \right) + c \\
&= \frac{1}{\sqrt{b-a}} \sin^{-1} \left(\tan x \sqrt{\frac{b-a}{a}} \right) + c
\end{aligned}$$

- Q27. Area enclosed by parabola $ay = 3(a^2 - x^2)$ and x-axis is 64, $a > 0$, let $y = f(x)$ be a real valued function satisfying $x \frac{dy}{dx} = x^2 + y - 2$, $f(1) = a - 3$, then :
- (A) $f(x)$ is minimum at $x = 1$ (B) $f(x)$ is maximum at $x = 1$ (C) $f(3) = 5$ (D) $f(2) = 3$

Ans. A, C

Sol. $\frac{xdy - ydx}{x^2} = \frac{x^2 - 2}{x^2} dx$
 $\int d\left(\frac{y}{x}\right) = \int \left(1 - \frac{2}{x^2}\right) dx$
 $y = x^2 - 2x + 2 \quad (\because f(1) = 1)$

- Q28. Let $f(x)$ be a polynomial function of degree 3 where $a < b < c$ and $f(a) = f(b) = f(c)$. If the graph of $f(x)$ is as shown, which of the following statements are INCORRECT ? (Where $c > |a|$) 

(A) $\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx$ (B) $\int_a^c f(x)dx < 0$ (C) $\int_a^b f(x)dx < \int_b^c f(x)dx$
 (D) $\frac{1}{b-a} \int_a^b f(x)dx > \frac{1}{c-b} \int_b^c f(x)dx$

Ans. B, C, D

Sol. $f(x) = -(x-a)(x-b)(x+c) = (x-a)(x)(x+c) [\because b=0]$

clearly option (A) is correct

(B) $\int_a^c f(x)dx = \int_a^b f(x)dx + \int_b^c f(x)dx > 0$ (from graph) which is incorrect

(C) $\int_a^b f(x)dx < 0$ & $\int_b^c f(x)dx < 0$ but second term is large negative value so option (c) is incorrect

(D) clearly D is incorrect.

- Q29. Let $f : R \rightarrow R$ is defined by $f(x) = x + \sin x$ & $\int_0^\pi f^{-1}(x)dx = I$, then :

(A) $I > \int_0^1 \frac{dx}{1+x^3}$ (B) $I < \int_0^1 e^{x^2} dx$ (C) $2 < I < 3$ (D) $\frac{\pi}{4} < I < \frac{\pi}{2}$

Ans. A, C

Sol. $I = \int_0^\pi f^{-1}(x)dx$

Let $x = f(t) \Rightarrow dx = f'(t)dt$

$\Rightarrow I = \int_0^\pi t f'(t)dt = \frac{\pi^2}{2} - 2 \approx 2.89$

Obviously $\int_0^1 \frac{dx}{1+x^3} < \int_0^1 dx$

$\Rightarrow \int_0^1 \frac{dx}{1+x^3} < 1 \Rightarrow$ (A) is true

- Q30. Let $f : R \rightarrow R$ be a real valued differentiable function satisfying the integral equation

$1 + \int_1^x 2t^2 f(t)dt = x^2 + 2 \int_x^1 f(t)dt$, then which of the following is (are) true :

(A) $f(x)$ has exactly one asymptote (B) $f(x)$ has neither injective nor surjective

(C) Distance between two horizontal tangents of $f(x)$ is 1

(D) If the equation $f(|x|) = k$ has four distinct solutions, then range of k is equal to $\left(0, \frac{1}{2}\right]$

Ans. A, B, C

Sol. Given equation $1 + \int_1^x 2t^2 f(t) dt = x^2 + 2 \int_x^1 f(t) dt$

diff. w.r.t. x

$$2x^2 f(x) = 2x + (0 - 2f(x))$$

$$\Rightarrow f(x) = \frac{x}{1+x^2}$$

clearly $y = \frac{1}{2}$ is only the horizontal tangent of $f(|x|)$

and $f(|x|) = k$ has four solutions for $k \in \left(0, \frac{1}{2}\right)$

Q31. If the probability that randomly selected point inside area bounded by parabola $y^2 = 8x$ and its latus rectum lies inside the ellipse $\frac{x^2}{2} + \frac{y^2}{16} = 1$ is $\frac{\sqrt{2}(a\pi + b)}{32}$ then. The points of extremum of

$$\int_0^{x^2} \left(\frac{t^2 - (a+b)t + 4}{2 + e^t} \right) dt \text{ are :}$$

(A) $x = -2$ (B) $x = 1$ (C) $x = 0$ (D) $x = -1$

Ans. A, B, C, D

Sol. Let $f(x) = \int_0^{x^2} \left(\frac{t^2 - 5t + 4}{2 + e^t} \right) dt$

$$f'(x) = \left(\frac{x^4 - 5x^2 + 4}{2 + e^{x^2}} \right) \times 2x$$

For extremum $f'(x) = 0$

Q32. Let $b > a > 0$, $I_1 = \int_a^{\frac{a+b}{2}} \left(\sqrt{\frac{b-x}{x-a}} - \sqrt{\frac{x-a}{b-x}} \right) dx$ and $I_2 = \int_{\frac{a+b}{2}}^b \left(\sqrt{\frac{x-a}{b-x}} - \sqrt{\frac{b-x}{x-a}} \right) dx$,

then :

(A) $I_1 = I_2$ (B) $I_1 > I_2$ (C) $I_1 < I_2$ (D) $I_1 + I_2 = 10$

Ans. A, D

Sol. $I_1 = \int_a^{\frac{a+b}{2}} \frac{(a+b) - 2x}{\sqrt{(x-a)(b-x)}} dx$

$$= 2\sqrt{(x-a)(b-x)} \Big|_a^{\frac{a+b}{2}}$$

$$= (b-a)$$

$$I_2 = \int_{\frac{a+b}{2}}^b \frac{2x - (a+b)}{\sqrt{(x-a)(b-x)}} dx$$

$$= -2\sqrt{(x-a)(b-x)} \Big|_{\frac{a+b}{2}}^b$$

$$= b-a$$

Hence $\therefore I_1 = I_2 = b-a$

Q33. Let $f(x-3) = f(x+3)$ for all $x \in R$ and $f(x) = \begin{cases} x & ; 0 \leq x < 3 \\ 6-x & ; 3 \leq x < 6 \end{cases}$, then :

(A) $\int_0^{36} f(x)dx = 54$ (B) $\int_0^{48} f(x)dx = 64$ (C) $\int_0^{54} f(x)dx = 81$ (D) $\int_0^{18} f(x)dx = 9$

Ans. A, C

Sol. $f(x)$ is periodic with period '6'

$$\int_0^6 f(x)dx = \frac{1}{2} \times 6 \times 3 = 9$$

$$\int_0^{36} f(x)dx = 6 \times 9 = 54$$

$$\int_0^{48} f(x)dx = 8 \times 9 = 72$$

$$\int_0^{54} f(x)dx = 9 \times 9 = 81$$

$$\int_0^{18} f(x)dx = 3 \times 9 = 27$$

Q34. The integral curve of the equation $(1-x^2)\frac{dy}{dx} + xy = x$ is :

(A) a conic whose centre is (0, 1) (B) a conic, length of whose one axis is 2

(C) an ellipse if $|x| < 1$ (D) a hyperbola if $|x| > 1$

Ans. A, B, C, D

Sol. -

Q35. If $\int \sqrt{\sec^2 x + 3} dx = \ln \left| f(x) + \sqrt{4 - f^2(x)} \right| + \sqrt{3} \sin^{-1} \left(\frac{k \cdot g(x)}{2} \right) + C$, where C is an integration constant, then :

(A) value of k is equal to $\sqrt{3}$ (B) $g(x) = \frac{f(x)}{\sqrt{1 + f^2(x)}}$ (C) $g^2(x) + \left(\sqrt{1 + f^2(x)} \right)^{-2} = 1$

(D) $g^2(x)f'(x) = f'(x) - g^2(x)$

Ans. A, C, D

Sol.
$$\begin{aligned} \int \frac{\sec^2 x + 3}{\sqrt{\sec^2 x + 3}} dx &= \int \frac{\sec^2 x dx}{\sqrt{4 + \tan^2 x}} + \int \frac{3 \cos x}{\sqrt{4 - 3 \sin^2 x}} dx \\ &= \int \frac{dt}{\sqrt{4 + t^2}} + \int \frac{\sqrt{3} d\theta}{\sqrt{4 - \theta^2}} \\ &= \ln \left| t + \sqrt{4 + t^2} \right| + \sqrt{3} \sin^{-1} \left(\frac{\theta}{2} \right) + C \\ &= \ln \left| \tan x + \sqrt{4 + \tan^2 x} \right| + \sqrt{3} \sin^{-1} \left(\frac{\sqrt{3} \sin x}{2} \right) + C \\ f(x) &= \tan x, g(x) = \sin x, k = \sqrt{3} \end{aligned}$$

Q36. In a question a student was given to find the derivative of the product of two functions f and g. The student by mistake thought $(fg)' = f'g'$ for his question $f(x) = x^3$ and he got the correct answer. Given that $g(4) = 1$.

Which of the following is/are correct possibilities (where $g(x) > 0$) :

(A) $g(x)$ is decreasing in $(-\infty, 3)$ (B) $g(x)$ is increasing in $(3, +\infty)$

(C) $g(x)$ is decreasing in $(3, +\infty)$ (D) $g(x)$ is increasing in $(-\infty, 3)$

Ans. C, D

Sol. $g'(x) = \frac{-3}{x-3}g(x)$
 $\Rightarrow \ln g(x) = -3 \ln |x-3| + c, c = 0$ as $g(4) = 1$
 $g(x) = \begin{cases} \frac{1}{(x-3)^3} & x > 3 \\ \frac{1}{(x-3)^3} & x < 3 \end{cases}$

Q37. The value of the definite integral $\int_{-\infty}^a \frac{(\sin^{-1} e^x + \sec^{-1} e^{-x})}{(\tan^{-1} e^a + \tan^{-1} e^{-x})(e^x + e^{-x})} dx$ ($a \in R$) is:

(A) Independent of a (B) dependent on a (C) $\frac{\pi}{2} \ln 2$ (D) $\frac{\pi}{2} \ln (2 \tan^{-1} e^a)$

Ans. A, B

Sol. We have $I = \int_{-\infty}^a \frac{(\sin^{-1} e^x + \cos^{-1} e^x)}{(\tan^{-1} e^a + \tan^{-1} e^{-x})} \left(\frac{e^x}{e^{2x} + 1} \right) dx$
 $= \frac{\pi}{2} \int_{-\infty}^a \frac{1}{(\tan^{-1} e^a + \tan^{-1} e^{-x})} \left(\frac{e^x}{e^{2x} + 1} \right) dx$
Put $\tan^{-1} e^x = t \Rightarrow \frac{e^x}{e^{2x} + 1} dx = dt$
 $I = \frac{\pi}{2} \int_0^{\tan^{-1} e^a} \frac{dt}{(t + \tan^{-1} e^a)}$
 $= \frac{\pi}{2} [\ln (t + \tan^{-1} e^a)]_0^{\tan^{-1} e^a}$
 $= \frac{\pi}{2} [\ln (2 \tan^{-1} e^a) - \ln (\tan^{-1} e^a)]$
 $= \frac{\pi}{2} \ln 2$

Q38. The value of $I = \int_{-1}^1 \left(A(\sin^{-1} x) + B \left(\frac{e^{2x} - 1}{e^{2x} + 1} \right) + C[x^2] + D \sin |x| \right) dx$ is independent of {where A, B, C, D $\in R$ and $[.]$ denotes greatest integer function} :

(A) A (B) B (C) C (D) D

Ans. A, B, C

Sol. Sol.
 $\sin^{-1} x, \frac{e^{2x} - 1}{e^{2x} + 1}$ are odd functions
and $x^2 \in [0, 1) \forall x \in (-1, 1)$

Q39. Let $f(x) = \int_1^x \frac{dt}{2+t^4}$, $f(2)$ satisfies inequality :

(A) $0 < f(2) < \frac{1}{2}$ (B) $\frac{1}{18} < f(2) < \frac{1}{3}$ (C) $f(2) > \frac{1}{3}$ (D) $f(2) > 1$

Ans. A, B

Sol. $f'(x) = \frac{1}{2+x^4}$

By LMVT $f'(c) = \frac{f(2) - f(1)}{2-1}$ for some $c \in (1, 2)$

⚠ Possibly misplaced — please review:

$$f'(c) = \frac{1}{2+c^4} \text{ as } f(1) = 0 \Rightarrow 1 < c < 2$$

$$3 < 2+c^4 < 18 \Rightarrow f(2) < \frac{1}{3}$$

Q40. If $I_1 = \int_1^e \frac{\ln(x+1)}{x^2} dx$, $I_2 = \int_0^1 \frac{e^x}{(x+1)^2} dx$ then :

(A) $I_1 > 1$ (B) $I_1 < 1$ (C) $I_2 < \ln 2$ (D) $I_1 > 1, I_2 > \ln 2$

Ans. B, D

Sol. $\therefore \ln(x+1) < x$

$$\Rightarrow \frac{\ln(x+1)}{x^2} < \frac{1}{x} \Rightarrow I_1 < 1$$

Now $x+1 < e^x \Rightarrow \frac{e^x}{(x+1)^2} > \frac{1}{x+1}$
 $\Rightarrow I_2 > \ln 2$

Q41. Let $I_n = \int_0^{\pi/2} \frac{\sec^2 x}{(\sec x + \tan x)^n} dx$; $n > 1$ then which of the following statement is/are correct :

(A) $I_3 = \frac{3}{8}$ (B) $I_2 + I_4 = \frac{14}{15}$ (C) $I_3 + I_5 = \frac{5}{12}$ (D) $I_2 + I_3 = \frac{25}{24}$

Ans. A, B, D

Sol. $I_n = \int_0^{\pi/2} \sec x \left(\frac{\sec^2 x + \sec x \tan x}{(\sec x + \tan x)^{n+1}} \right) dx$

$$I_n = \left(\frac{\sec x}{-n(\sec x + \tan x)^n} \right)_0^{\pi/2} + \frac{1}{n} \int_0^{\pi/2} \frac{\sec x \cdot \tan x}{(\sec x + \tan x)^n} dx$$

$$= \frac{1}{n} + \frac{1}{n} \left[\int_0^{\pi/2} \frac{\sec x \tan x + \sec^2 x}{(\sec x + \tan x)^n} dx - \int_0^{\pi/2} \frac{\sec^2 x dx}{(\sec x + \tan x)^n} \right]$$

$$\begin{aligned}
I_n &= \frac{1}{n} + \frac{1}{n} \left[\frac{1}{(-n+1)(\sec x + \tan x)^{n-1}} \right]_0^{\pi/2} - \frac{1}{n} I_n \\
&\Rightarrow \left(1 + \frac{1}{n} \right) I_n \\
&= \frac{1}{n} + \frac{1}{n(n-1)} \\
&= \frac{1}{n-1} \\
&\Rightarrow I_n = \frac{n}{n^2-1} \\
&= \frac{1}{n} \left[\frac{1}{(-n+1)(\sec x + \tan x)^{n-1}} \right]_0^{\pi/2} - \frac{1}{n} I_n \\
&\Rightarrow \left(1 + \frac{1}{n} \right) I_n \\
&= \frac{1}{n} + \frac{1}{n(n-1)} \\
&= \frac{1}{n-1} \\
&\Rightarrow I_n = \frac{n}{n^2-1}
\end{aligned}$$

Q42. Choose the correct option(s), if $\int_{\frac{3}{2}}^x \frac{x^2+3}{x^4+x^3-x^2-3x+9} dx = f(x)$

- (A) $f(2) = \frac{2}{\sqrt{19}} \tan^{-1} \frac{2}{\sqrt{19}}$ (B) $f(2) = \frac{2}{\sqrt{19}} \tan^{-1} \frac{4}{\sqrt{19}}$ (C) $f(3) = \frac{2}{\sqrt{19}} \tan^{-1} \frac{5}{\sqrt{19}}$
(D) $f(3) = \frac{2}{\sqrt{19}} \tan^{-1} \frac{3}{\sqrt{19}}$

Ans. A, C

Sol. -

Q43. Let $y = f(x)$ be a real valued function satisfying $x \frac{dy}{dx} = x^2 + y - 2$, $f(1) = 1$, then :

- (A) $f(x)$ is minimum at $x = 1$ (B) $f(x)$ is maximum at $x = 1$ (C) $f(3) = 5$ (D) $f(2) = 3$

Ans. A, C

Sol. $x \frac{dy}{dx} = x^2 + y - 2$

$$x \frac{dy}{dx} - y = x^2 - 2$$

$$\frac{x \frac{dy}{dx} - y}{x^2} = 1 - \frac{2}{x^2}$$

$$\frac{d}{dx} \left(\frac{y}{x} \right) = 1 - \frac{2}{x^2}$$

Integrating both sides:

$$\frac{y}{x} = x + \frac{2}{x} + C$$

$$y = x^2 + 2 + Cx$$

$$\text{Given } f(1) = 1$$

$$1 = 1^2 + 2 + C(1)$$

$$1 = 1 + 2 + C$$

$$C = -2$$

$$\text{So, } f(x) = x^2 - 2x + 2$$

To find minimum/maximum, find $f'(x)$:

$$f'(x) = 2x - 2$$

$$\text{Set } f'(x) = 0:$$

$$2x - 2 = 0 \Rightarrow x = 1$$

Now find $f''(x)$:

$$f''(x) = 2$$

Since $f''(1) = 2 > 0$, $f(x)$ has a local minimum at $x = 1$.

So, option (A) is correct.

For option (C):

$$f(3) = (3)^2 - 2(3) + 2 = 9 - 6 + 2 = 5$$

So, option (C) is correct.

For option (D):

$$f(2) = (2)^2 - 2(2) + 2 = 4 - 4 + 2 = 2$$

So, option (D) is incorrect.

Therefore, options (A) and (C) are correct.

$$\int d\left(\frac{y}{x}\right) = \int \left(1 - \frac{2}{x^2}\right) dx$$

$$y = x^2 - 2x + 2 \quad (\because f(1) = 1)$$

$$\frac{dy}{dx} = 2x - 2 = 0 \Rightarrow x = 1$$

$$\frac{d^2y}{dx^2} > 0 \text{ at } x = 1, f(x) \text{ will ne minimum.}$$

Q44. Solution of the differential equation : $\frac{x + y \frac{dy}{dx}}{y - x \frac{dy}{dx}} = \frac{x \sin^2(x^2 + y^2)}{y^3}$

(A) $-\cot(x^2 + y^2) = \left(\frac{x}{y}\right)^2 + c$ (B) $\frac{y^2}{x^2 + y^2} c = -\tan(x^2 + y^2)$

(C) $-\cot(x^2 + y^2) = \left(\frac{y}{x}\right)^2 + c$ (D) $\frac{y^2 + x^2 c}{x^2} = -\tan(x^2 + y^2)$

Ans. A, B

Sol. The given differential equation can be written as

$$\frac{x dx + y dy}{y dx - x dy} = \frac{x \sin^2(x^2 + y^2)}{y^3}$$

$$\Rightarrow \frac{2x dx + 2y dy}{y dx - x dy}$$

$$= \frac{2x \sin^2(x^2 + y^2)}{y^3}$$

$$\Rightarrow \frac{d(x^2 + y^2)}{\sin^2(x^2 + y^2)}$$

$$= \frac{2x}{y^3} (y dx - x dy)$$

$$\Rightarrow \operatorname{cosec}^2(x^2 + y^2) d(x^2 + y^2)$$

$$= 2 \left(\frac{x}{y}\right) d\left(\frac{x}{y}\right)$$

On integrating, we get

$$\int \operatorname{cosec}^2(x^2 + y^2) d(x^2 + y^2) = 2 \int \left(\frac{x}{y}\right) d\left(\frac{x}{y}\right) \\ \Rightarrow -\cot(x^2 + y^2) = \left(\frac{x}{y}\right)^2 + C, \text{ which is the required solution.}$$

Q45. The area bounded by $\min(|x|, |y|) = 2015$ and $\max(|x|, |y|) = 2016$ is less than or equal to :

- (A) 4 sq unit (B) 8 sq unit (C) 2 sq unit (D) 1 sq unit

Ans. A, B

Sol. Required area $= 4 \times (1 \times 1) = 4$ sq unit

Q46. Solution of the differential equation $(2 + 2x^2\sqrt{y})dx + (x^2\sqrt{y} + 2)x dy = 0$ is/are :

- (A) $xy(x^2\sqrt{y} + 5) = c$ (B) $xy(x^2\sqrt{y} + 3) = c$ (C) $xy(y^2\sqrt{x} + 3) = c$
(D) $xy(y^2\sqrt{x} + 5) = c$

Ans. B

Sol. $2ydx + 2xdy + (2x^3y^{3/2}dx + x^3y^{1/2}dy) = 0$

$$2d(xy) + \frac{2}{3}d(x^3y^{3/2}) = 0$$

On integrating

$$xy + \frac{1}{3}(x^3 \cdot y^{3/2}) = 0 \\ \Rightarrow xy(3 + x^2\sqrt{y}) = 0$$

Paragraph

Let $y = f(x)$ satisfy the equation $f(x) = (e^{-x} + e^x) \cos x - 2x - \int_0^x (x-t)f'(t)dt$. Then,

Q47. $f(x)$ satisfies the differential equation :

- (A) $\frac{dy}{dx} + y = e^x(\cos x - \sin x) - e^{-x}(\cos x + \sin x)$
(B) $\frac{dy}{dx} = y + e^x(\cos x + \sin x) + e^{-x}(\cos x - \sin x)$
(C) $\frac{dy}{dx} + y = e^x(\cos x + \sin x) - e^{-x}(\cos x + \sin x)$ (D) None of these

Ans. A

Sol. -

Q48. $f(0) + f''(0) =$:

- (A) -1 (B) 2 (C) 0 (D) 1

Ans. C

Sol. -

Q49. $f(x)$ as a function of x equals :

- (A) $e^{-x}(\cos x - \sin x) + \frac{e^x}{5}(3 \cos x + \sin x) + \frac{2}{5}e^{-x}$
 (B) $e^{-x}(\cos x + \sin x) + \frac{e^x}{5}(3 \cos x - \sin x) - \frac{2}{5}e^{-x}$
 (C) $e^{-x}(\cos x - \sin x) + \frac{e^x}{5}(3 \cos x - \sin x) + \frac{2}{5}e^{-x}$ (D) None of these

Ans. C

Sol. -

Paragraph

Consider two differentiable functions $f(x)$, $g(x)$ satisfying $6 \int f(x)g(x)dx = x^6 + 3x^4 + 3x^2 + c$ & $2 \int \frac{g(x)dx}{f(x)} = x^2 + c$, where $f(x) > 0 \forall x \in R$.

Q50. $f'(1) + g'(2)$ is equal to :

- (A) 15 (B) 14 (C) 13 (D) 12

Ans. A

Sol. Differentiating both sides we get

$$g(x) \cdot f(x) = (x^2 + 1)(x^3 + x)$$

$$\Rightarrow \frac{g(x)}{f(x)} = x$$

$$\Rightarrow g(x) = x^3 + x$$

$$f(x) = x^2 + 1$$

Q51. $\int (f(x) + g(x))dx$ is equal to :

- (A) $\frac{x^4}{4} + \frac{x^2}{2} + x + c$ (B) $\frac{x^4}{4} + \frac{x^2}{3} + x^2 + 2x + c$ (C) $\frac{x^4}{4} + \frac{x^3}{3} + \frac{x^2}{2} + x + c$
 (D) $\frac{x^4}{4} + \frac{x^3}{3} + c$

Ans. C

Sol. -

Paragraph

Let $y'(x)\phi(x) - y(x)\phi'(x) + y^2(x) = 0$, $y(1) = 1$, $x \in R$, $\phi(x)$ and $y(x)$ are non constant differentiable function on R with $\phi(1) = 0$ and $\phi(2) = 4$.

Q52. The value of $y(2)$ is equal to :

- (A) 0 (B) 1 (C) 2 (D) 4

Ans. D

Sol. $y'(x)\phi(x) - y(x)\phi'(x) + y^2(x) = 0$
 $\frac{d}{dx} \left(\frac{\phi(x)}{y(x)} \right) = 1$
 $\Rightarrow \frac{\phi(x)}{y(x)} = x - 1 \quad \{\because \phi(1) = 0\}$
 $\therefore \text{for } x = 2 \Rightarrow y(2) = \phi(2) = 4$

Q53. The value of $\int_1^2 \left(\frac{\phi(x)}{y(x)} \right) \frac{1}{x^2 \sqrt{x^2 + \left(\frac{\phi(x)}{y(x)} \right)^2}} dx$ is equal to :

- (A) $\sqrt{5}$ (B) $\frac{\sqrt{5}}{2} - 1$ (C) $\sqrt{5} - 1$ (D) $\sqrt{5} - 2$

Ans. B

Sol. $\int_1^2 \frac{x-1}{x^2 \sqrt{x^2 + (x-1)^2}} dx = \int_1^2 \frac{x \left(1 - \frac{1}{x}\right)}{x^2 \sqrt{x^2 \left(1 + \left(\frac{1}{x} - 1\right)^2\right)}} dx$
 $= \int_0^{1/2} \frac{t}{\sqrt{1+t^2}} dt$ where $t = \frac{1}{x} - 1$
 $= 1 - \frac{1}{x}$
 $= [\sqrt{1+t^2}]_0^{1/2}$
 $= \sqrt{1 + \frac{1}{4}} - 1$
 $= \frac{\sqrt{5}}{2} - 1$
 $= \int_0^{1/2} \frac{t}{\sqrt{1+t^2}} dt$ where $t = \frac{1}{x} - 1$
 $= 1 - \frac{1}{x}$
 $= [\sqrt{1+t^2}]_0^{1/2}$
 $= \sqrt{1 + \frac{1}{4}} - 1$
 $= \frac{\sqrt{5}}{2} - 1$

Q54. Which one of the following statement is correct for function $g(x) = \frac{\phi(x)}{y(x)}$:

- (A) $g(x)$ is decreasing on $\mathbb{R}$ (B) Area bounded by $g(x)$ and coordinate axes is $\frac{1}{4}$
 (C) $g(x)$ has minima but no maxima (D) $|g(x)|$ is continuous but not differentiable on $\mathbb{R}$

Ans. D

Sol. $g(x) = \frac{\phi(x)}{y(x)} = x - 1$

Paragraph

The tangent at point P of a curve $y = f(x)$ meets the y-axis at A, x-axis at B and the line parallel to y-axis through P meets x-axis at C such that area of ΔPBC is equal to the ratio of ordinate to abscissae of point P where $f(1) = 1$ and $f(3) = e^2$.

Q55. If the area bounded by the curve $y = f(x)$ line $x + y = 2$ and y-axis in the I^{st} quadrant is A, then :

- (A) $\frac{1}{2} < A < 1$ (B) $1 < A < 2$ (C) $2 < A < 3$ (D) $3 < A < 4$

Ans. A

Sol. -

Paragraph

Let $f(x)$ be a continuous function such that $f(x) = \int_0^x (f(t) \cos t - \cos(t-x)) dt$.

Q56. Number of integers in range of $f(x)$ is :

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol.
$$f(x) = \int_0^x f(t) \cos t dt - \int_0^x \cos(t-x) dt$$

$$f(x) = \int_0^x f(t) \cos t dt - \int_0^x \cos t dt$$

$$f'(x) = f(x) \cos x - \cos x$$

$$\Rightarrow \frac{f'(x)}{f(x) - 1} = \cos x$$

Integrating both sides w.r.t. x, we get

$$\ln |f(x) - 1| = \sin x + C$$

$$\text{Now } f(0) = 0$$

$$\Rightarrow c = 0 \Rightarrow |f(x) - 1| = e^{\sin x}$$

$$f(x) = 1 \pm e^{\sin x}$$

$$\text{But } f(0) = 0 \Rightarrow f(x) = 1 - e^{\sin x}$$

Now $f(x)$ is periodic

$$\text{Also } f(x) = 0$$

$$\Rightarrow e^{\sin x} = 1$$

$$\Rightarrow \sin x = 0$$

$$\Rightarrow x = 0, \pi, 2\pi$$

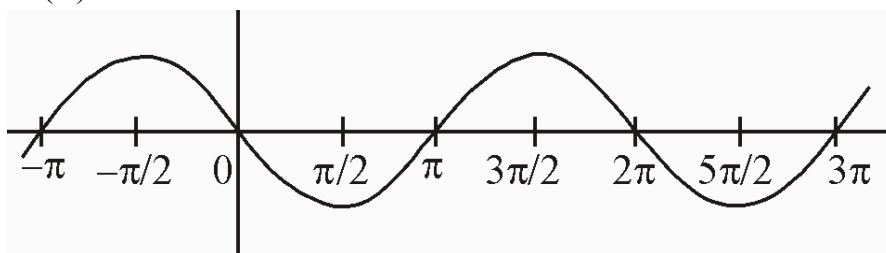
Also suppose $f(x) = 0$ has two roots α & β

$$\therefore f(\alpha) = f(\beta)$$

$\therefore$ Using Rolle's theorem the equation $f'(x) = 0$ has at least one root in $x \in (\alpha, \beta)$

But $f(x)$ is non-monotonic

$\therefore$ (D) is false.



$$\therefore \text{Range of } f(x) \text{ is } \left[1 - e, 1 - \frac{1}{e}\right]$$

$\therefore$ 2 integers.

Q57. Let $f(x)$ and $g(x)$ are two differentiable function on $\mathbb{R}$ satisfying $f(x) = \frac{x^3}{2} + 1 - x \int_0^x g(t)dt$ and $g(x) = x - \int_0^1 f(t)dt$. If minimum distance between $f(x)$ and $g(x)$ is d , then $\sqrt{2}d$ is _____.

Ans. 2.33

Sol. Sol.

$$g(x) = x - k \text{ where } k = \int_0^1 f(t)dt$$

$$f(x) = \frac{x^3}{2} + 1 - x \int_0^x g(t)dt \quad \{f(0) = 1\}$$

differentiating w.r.t. x

$$f'(x) = \frac{3x^2}{2} - xg(x) - \int_0^x g(t)dt$$

$$= \frac{3x^2}{2} - x(x - k) - \int_0^x (t - k)dt$$

$$= \frac{x^2}{2} + xk - \left(\frac{t^2}{2} - kt\right)_0^x = 2kx$$

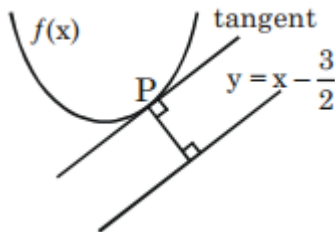
$$f(x) = kx^2 + c \Rightarrow f(x) = kx^2 + 1 \{ \because f(0) = 1 \}$$

$$\Rightarrow k = \int_0^1 (kt^2 + 1)dt$$

$$k = \left(\frac{kt^3}{3} + t\right)_0^1$$

$$k = \frac{3}{2} \Rightarrow f(x) = \frac{3}{2}x^2 + 1, g(x) = x - \frac{3}{2}$$

$\therefore$ Minimum distance between $f(x)$ and $g(x)$ is along common normal



$$f'(x) = 1$$

$$\Rightarrow x$$

$$= \frac{1}{3}, y$$

$$= \frac{7}{6}$$

$$\Rightarrow P\left(\frac{1}{3}, \frac{7}{6}\right)$$

$$\begin{aligned} \Rightarrow \text{Minimum distance} &= \left(\text{distance of } P \text{ from } y = x - \frac{3}{2}\right) \\ &= \frac{7}{3\sqrt{2}} \end{aligned}$$

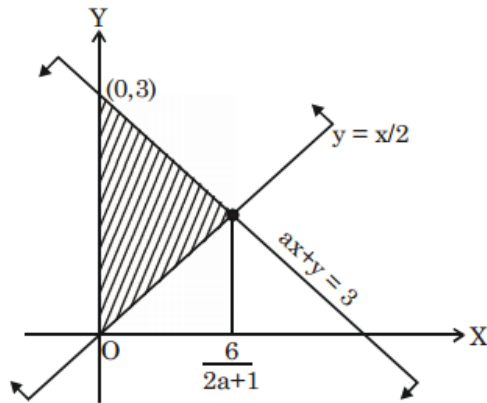
Q58. Let $a > -\frac{1}{2}$ and area bounded by region $x \geq 0, 2y - x \geq 0, ax + y - 3 \leq 0$ is equal to 3, then value of a is _____.

Ans. 1.00

Sol. $2a + 1 > 0$

solving $y = 3 - ax, 2y = x$

$$\Rightarrow x = \frac{6}{2a+1}$$



$$\text{Area} = \frac{1}{2}(3) \left(\frac{6}{2a+1} \right) = 3$$

$$a = 1$$

Q59. Let $y = f(x)$ is the solution of the differential equation $y \frac{d^2y}{dx^2} - 2 \left(\frac{dy}{dx} \right)^2 = 0, f(1) = 1$ and $f(0) = 2$.

If S is area bounded by curves $y = f(x), y = 2\sqrt{\frac{2}{f(x)}}$ and line $f(x) = \frac{2}{e}$ and x -axis, then $\frac{S}{5}$ is equal to _____.

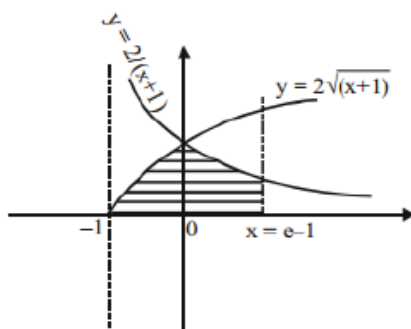
Ans. 0.66

Sol. $y \frac{d^2y}{dx^2} = 2 \left(\frac{dy}{dx} \right)^2$

$$\frac{\frac{d^2y}{dx^2}}{\left(\frac{dy}{dx} \right)} = \frac{2}{y} \frac{dy}{dx}$$

on integrate we get $\frac{dy}{dx} = ky^2$

$$\frac{dy}{y^2} = k dx$$



$$\therefore y = -\frac{1}{kx+c}$$

$$\text{given } f(1) = 1 \Rightarrow c = -1 - k$$

given $f(0) = 2$

$$\therefore 2 = -\frac{1}{c}$$

$$c = -\frac{1}{2}$$

$$\therefore f(x) = y = \frac{2}{x+1}$$

$\Rightarrow$ other curve is

$$y = 2\sqrt{\frac{2}{f(x)}}$$

$$= 2\sqrt{x+1}$$

$$f(x) = \frac{2}{e}$$

$$\Rightarrow \frac{2}{x+1}$$

and line is

$$= \frac{2}{e}$$

$$\Rightarrow x$$

$$= e-1$$

Now, required area is

$$S = \int_{-1}^0 2\sqrt{x+1} dx + \int_0^{e-1} \frac{2}{x+1} dx$$

$$= \left[\frac{4}{3}(x+1)^{\frac{3}{2}} \right]_{-1}^0 + [2 \ln(x+1)]_0^{e-1}$$

$$= \frac{4}{3} + 2 = \frac{10}{3}$$

Q60. Let f be a differentiable function satisfying $f'(x) = f^2(x)$ and $f(0) = -1$. If

$$\frac{\sqrt{2}}{\pi} \int_0^1 \left(\frac{(1-x)}{f(x)(x^4+1)} + \frac{2}{x^4+1} \right) dx = \frac{p}{q} \text{ where } p, q \in N \text{ then find least value of } (p+q)$$

_____.

Ans. 3

Sol. $\frac{dy}{dx} = y^2$

$$\frac{dx}{y^2} = dx$$

$$-\frac{1}{y} = x + c$$

$$-\frac{1}{-1} = 0 + c$$

$$c = 1$$

$$y = \frac{-1}{x+1}$$

Put in Integration

$$\frac{\sqrt{2}}{\pi} \int_0^1 \left(\frac{x^2-1}{x^4+1} + \frac{2}{x^2+1} \right) dx = \frac{\sqrt{2}}{\pi} \int_0^1 \frac{x^2+1}{x^4+1} dx$$

$$= \frac{\sqrt{2}}{\pi} \int_0^1 \frac{1 + \frac{1}{x^2}}{\left(x - \frac{1}{x}\right)^2 + 2} dx$$

$$x - \frac{1}{x} = t$$

$$\frac{\sqrt{2}}{\pi} \int_{-\infty}^0 \frac{dt}{t^2+2}$$

$$\frac{\sqrt{2}}{\pi} \frac{1}{\sqrt{2}} \left[\tan^{-1} \frac{t}{\sqrt{2}} \right]_{-\infty}^0$$

$$\frac{1}{\pi} \left(0 + \frac{\pi}{2} \right) = \frac{1}{2}$$

Q61. Let $f : R \rightarrow R$, be a continuous function with $\int_0^1 f(x)f'(x)dx = 0$ and $\int_0^1 f^2(x)f'(x)dx = 18$. If the value of $\int_0^1 f^4(x)f'(x)dx = \frac{p}{q}$ where p and q are relatively prime positive integers, find $(p + q)$

Ans. 491

Sol. $f : R \rightarrow R$

$$\int_0^1 f(x)f'(x)dx = 0 \text{ and } \int_0^1 f^2(x)f'(x)dx = 18$$

$$\int_0^1 f^2(x)f'(x)dx = [f^2(x)f(x)]_0^1 - \int_0^1 2f(x) \cdot f'(x) \cdot f(x)dx$$

$$\int_0^1 f^2(x)f'(x)dx = f^3(1) - f^3(0) - 2 \int_0^1 f^2(x)f'(x)dx$$

$$3 \times 18 = f^3(1) - f^3(0)$$

$$f^3(1) - f^3(0) = 54 \quad (1)$$

$$\int_0^1 f(x)f'(x)dx = 0$$

$$\Rightarrow \int_0^1 f(x)f'(x)dx$$

$$= [f^2(x)]_0^1 - \int_0^1 f(x)f'(x)dx$$

$$f^2(1) - f^2(0) = 0$$

$$f(1) = f(0) \text{ or } f(1) = -f(0)$$

$$\therefore f^3(1) - f^3(0) = 54 \Rightarrow f(1) = -f(0)$$

$$2f^3(1) = 54$$

$$f^3(1) = 27$$

$$f(1) = 3 \Rightarrow f(0) = -3$$

$$\Rightarrow \int_0^1 f^4(x)f'(x)dx$$

$$= [f^4(x)f(x)]_0^1 - \int_0^1 4f^3(x)f'(x)f(x)dx$$

$$5 \int_0^1 f^4(x)f'(x)dx = f^5(1) - f^5(0)$$

$$= 2 \times (3)^5$$

$$\int_0^1 f^4(x)f'(x)dx = \frac{486}{5} = \frac{p}{q}$$

$$p + q = 491$$

Q62. A real number r such that the equation $|x + 2| + |x - 5| = r$ has infinitely many solutions in x. Find the area of the figure enclosed by the curve $5x^2 + 6xy + 2y^2 + rx + 6y + 6 = 0$.

Ans. 2

Sol. $r = 7, 5x^2 + 6xy + 2y^2 + 7x + 6y + 6 = 0$

$$y = \frac{-3(1+x) \pm \sqrt{(3-x)(x-1)}}{2}; (1 \leq x \leq 3)$$

$$\begin{aligned}\text{Area} &= \int_1^3 |y_1 - y_2| dx = \int_1^3 \sqrt{(3-x)(x-1)} dx \\ &= \frac{\pi(3-1)^2}{8} = \frac{4\pi}{8} = \frac{\pi}{2} \\ \text{So, } 3\lambda &= \frac{\pi}{2} \Rightarrow \lambda = \frac{\pi}{6} \\ c^2 + m^2 &= 1 + 1 = 2\end{aligned}$$

Q63. Consider $f(x) = x^2 - 4x + 3 \forall x \in R$. Area bounded by curve $y = |f(|x|)|$ between $1 \leq |x| \leq 3$ and x-axis is $\frac{2k}{3}$ sq. units. One of the diameters of the circle circumscribing the rectangle ABCD in order is $ky = x + 7$. If A & B are the points $(-3, 4)$ & $(5, 4)$ respectively, then find the area of the rectangle.

Ans. 32

Sol. Graph of $y = |f(|x|)|$

Required area

$$2. \int_1^3 (x^2 - 4x + 3) dx = \frac{8}{3} \text{ sq. units}$$



Q64. Let $I_n = \int_{-1}^1 |x| \left(1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^{2n}}{2n} \right) dx$. If $\lim_{n \rightarrow \infty} I_n$ can be expressed as rational $\frac{p}{q}$ in its lowest form, then find the value of $pq(p+q)$.

Ans. 30

Sol.

$$\begin{aligned}I_n &= \int_{-1}^1 |x| \left(1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^{2n}}{2n} \right) dx \\ &= 2 \left[\int_0^1 x \left(x + \frac{x^3}{2} + \frac{x^5}{4} + \dots + \frac{x^{2n+1}}{2n} \right) dx \right] \\ &= 2 \left[\frac{x^2}{1 \cdot 2} + \frac{x^4}{2 \cdot 4} + \frac{x^6}{4 \cdot 6} + \dots + \frac{x^{2n+2}}{2n(2n+2)} \right]_0^1 \\ &= 2 \left[\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 4} + \frac{1}{4 \cdot 6} + \dots + \frac{1}{2n(2n+2)} \right] \\ I_n &= 1 + \frac{1}{2} \left(\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)} \right) \\ &= 1 + \frac{1}{2} \left(1 - \frac{1}{n+1} \right) \\ \lim_{n \rightarrow \infty} I_n &= 1 + \frac{1}{2} = \frac{3}{2} = \frac{p}{q} \\ pq(p+q) &= 3 \times 2(5) = 30\end{aligned}$$

Q65. $\lim_{n \rightarrow \infty} \left(\frac{{}^{3n}C_n}{{}^{2n}C_n} \right)^{\frac{1}{n}}$ is equal to $\frac{a}{b}$, where a and b are co-prime number then $a + b$ is equal to _____.

Ans. 43

Sol.

$$\begin{aligned}
 y &= \lim_{n \rightarrow \infty} \left(\frac{3n!}{2n!n!} \cdot \frac{n!n!}{2n!} \right)^{\frac{1}{n}} \\
 y &= \lim_{n \rightarrow \infty} \left(\frac{(3n)!}{(2n)!(n)!} \right)^{\frac{1}{n}} \\
 &= \lim_{n \rightarrow \infty} \left(\frac{\{(3n)(3n-1)\dots(2n+1)\}\{(n)(n-1)\dots(1)\}}{\{(2n)(2n-1)\dots(n+1)\}\{(n)(n-1)\dots(1)\}\{(n)(n-1)\dots(1)\}\{(n)(n-1)\dots(1)\}} \right)^{\frac{1}{n}} \\
 &= \lim_{n \rightarrow \infty} \left(\prod_{r=0}^{n-1} \frac{3n-r}{2n-r} \right)^{\frac{1}{n}} \\
 \ln y &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{n-1} \ln \left(\frac{3n-r}{2n-r} \right) \\
 \ln y &= \int_0^1 \ln \left(\frac{3-x}{2-x} \right) dx \\
 \ln y &= \int_0^1 \ln(3-x) dx - \int_0^1 \ln(2-x) dx \\
 \ln y &= \int_0^1 \ln(x+2) dx - \int_0^1 \ln(x+1) dx \\
 \ln y &= [(x+2)\ln(x+2) - (x+2)]_0^1 - [(x+1)\ln(x+1) - (x+1)]_0^1 \\
 \ln y &= [(3\ln 3 - 3) - (2\ln 2 - 2)] - [(2\ln 2) - (1\ln 1 - 1)] \\
 \ln y &= 3\ln 3 - 4\ln 2 = \ln \frac{27}{16} \Rightarrow y = \frac{27}{16}
 \end{aligned}$$

Q66. The rate at which a substance cools in moving air is proportional to the difference between the temperatures of the substance and that of the air. If the temperature of the air is 290 K and the substance cools from 370K to 330K in 10 minutes, when will the temperature be 295 K _____.

Ans. 40

Sol. Let T be the temperature of the substance at time t.

$$\begin{aligned}
 \therefore -\frac{dT}{dt} &= k(T - 290) \\
 \text{or } \frac{dT}{dt} &= k(T - 290) \text{ (Negative sign because } \frac{dT}{dt} \text{ is rate of cooling)}
 \end{aligned}$$

$$\text{or } \int \frac{dT}{T - 290} = -k \int (dt)$$

Integrating the L.H.S. between the limits $T = 370$ to $T = 330$ and the R.H.S. between the limits $t = 0$ to $t = 10$, we get

$$\int_{370}^{330} \frac{dT}{T - 290} = -k \int_0^{10} dt$$

$$\text{or } \log 40 - \log 80 = -k \times 10$$

$$\log 2 = 10k$$

$$\text{or } k = \frac{\log 2}{10} \dots\dots\dots (2)$$

Now, integrating equation (1) between $T = 370$ and $T = 295$ and $t = 0$ and $t = 1$, we get

$$\int_{370}^{295} \frac{dT}{T - 290} = -k \int_0^1 dt$$

$$\text{or } \log(T - 290) = \frac{295}{370} kt$$

$$\text{or } r = \frac{\log 16}{k}$$

Hence, from equation (2), we get $\log t = \frac{\log 16}{k}$

Hence, from equation (2), we get $t = \frac{\log 16}{\log 2} \times 10 = 40$

I.t.. after 40 min

Q67. If the tangent at the point P on the circle $x^2 + y^2 + 6x + 6y = 2$ meets the straight line $5x - 2y + 6 = 0$ at a point Q on the y-axis, then the length of PQ is λ .

What constant interest rate is required if an initial deposit placed into an account the accrues interest compounded continuously is to double its value in $(\lambda + 1)$ years. $(\ln 2 = 0.6930)$ _____.

Ans. 11.55

Sol. -

Q68. The function $f(x) = \int_4^{x^2} \sqrt{9+t^2} dt$ is invertible. The value of $100(f^{-1})'(0)$ equal . _____.

Ans. 20

Sol. Ans. $\frac{1}{20}$

Sol. at $y = 0, x = 2$

$$f'(x) = \sqrt{9+x^4} \cdot 2x$$

$$\begin{aligned} \therefore g'(y) &= \frac{1}{f'(x)} \Big|_{x=2} \\ &= \frac{1}{2x\sqrt{9+x^4}} \\ &= \frac{1}{20} \end{aligned}$$

Q69. The set of points (x, y) in the plane satisfying $x^{2/5} + |y| = 1$ form a curve enclosing a region of area $\frac{p}{q}$ square units, where p and q are relatively prime positive integers. Find $p + q$.

Ans. 15

Sol.
$$= 4 \int_0^1 (1 - x^{2/5}) dx$$

$$\text{Ar} = 4 \left(x - \frac{5}{7} x^{7/5} \right) \Big|_0^1$$

$$= \frac{8}{7}$$

$$p = 8, q = 7$$

$$p + q = 15$$

Q70. Let A_n represents the area formed by the elements of set A, defined below.

$A = \{(x, y) \in R \times R : y \leq [x] + \sqrt{\{x\}}, y \geq 0, 0 \leq x \leq n, n \in N\}$, $[.]$ denotes the greatest integer function and $\{.\}$ denotes the fractional part function, then find the value of $A_3 + A_6$.

Ans. 24

Sol. $y \leq [x] + \sqrt{\{x\}}, y \geq 0, 0 \leq x \leq n, n \in N$

$$A_n = \int_0^1 \sqrt{x} dx + \int_1^2 (1 + \sqrt{x-1}) dx + \int_2^3 (2 + \sqrt{x-2}) dx + \dots + \int_{n-1}^n ((n-1) + \sqrt{x-(n-1)}) dx$$

$$\begin{aligned}
A_n &= \left[\frac{2}{3}(x)^{\frac{3}{2}} \right]_0^1 + \left[x + \frac{2}{3}(x-1)^{\frac{3}{2}} \right]_1^2 + \dots + \\
A_n &= \frac{2}{3} + \left[(2-1) + \frac{2}{3}(2-1)^{\frac{3}{2}} \right] + \left[(3-2) + \frac{2}{3}(3-2)^{\frac{3}{2}} \right] + \\
&\dots + (n-1)[n - (n-1)] + \frac{2}{3}(n - (n-1))^{\frac{3}{2}} \\
A_n &= \frac{2}{3} + \left(1 + \frac{2}{3} \right) + \left(2 + \frac{2}{3} \right) + \dots + \left[(n-1) + \frac{2}{3} \right] \\
A_n &= \frac{2}{3}(n) + \frac{(n-1)(n)}{2}
\end{aligned}$$

- Q71. If the two lines AB : $\left(\int_0^{2t} \left(\frac{\sin x}{x} + 1 \right) dx \right) x + y = 3t$ and AC : $2tx + y = 0$ intersect at a point A, then x-coordinate of point A as $t \rightarrow 0$, is equal to $\frac{p}{q}$ (p and q are in their lowest form). Find $(p + q)$.

Ans. 7

Sol.

$$\begin{aligned}
x_A &= \lim_{t \rightarrow 0} \frac{3t}{\int_0^{2t} \left(\frac{\sin x}{x} + 1 \right) dx} \\
\text{Sol.} \quad &= \lim_{t \rightarrow 0} \frac{3t}{\int_0^{2t} \frac{\sin x}{x} dx + \int_0^{2t} 1 dx} \\
&= \lim_{t \rightarrow 0} \frac{3t}{\int_0^{2t} \frac{\sin x}{x} dx + 2t} \\
&= \lim_{t \rightarrow 0} \frac{3}{\frac{\int_0^{2t} \frac{\sin x}{x} dx}{t} + 2}
\end{aligned}$$

Using L'Hopital's rule for $\lim_{t \rightarrow 0} \frac{\int_0^{2t} \frac{\sin x}{x} dx}{t}$

$$\begin{aligned}
&= \lim_{t \rightarrow 0} \frac{\frac{\sin(2t)}{2t} \cdot 2}{1} \\
&= \lim_{t \rightarrow 0} \frac{\sin 2t}{2t} \cdot 2 \\
&= 1 \cdot 2 \\
&= 2
\end{aligned}$$

$$\text{So, } x_A = \frac{3}{2+2} = \frac{3}{4}$$

$$p = 3, q = 4$$

$$p + q = 7$$

- Q72. Find the value of $3I$, where I is defined as $I = \sum_{r=0}^{10} \int_r^{r+4} |x-1-r||x-2-r||x-3-r| dx$

Ans. 165

Sol. -

- Q73. If $(\sin(x^2y) + 2x^2y \cos(x^2y))dx + (x^3 \cos(x^2y) + 1)dy = 0$, $y(0) = 0$, $y(\alpha) = 1$, then $|\alpha \sin(\alpha^2)|$ is

_____.

Ans. 1

Sol. -

- Q74. If $\int \sin 4x e^{\tan^2 x} dx = c - A \cos^4 x e^{\tan^2 x}$ then $A =$ _____.

Ans. 2

Sol.
$$I = \int \sin x \cos x \left(\frac{1 - \tan^2 x}{1 + \tan^2 x} \right) e^{\tan^2 x} dx = 4 \int \tan x \sec^2 x \cos^6 x (1 - \tan^2 x) e^{\tan^2 x}$$
$$t = \tan^2 x$$
$$\Rightarrow I$$
$$= 2 \int \frac{(1-t)e^t}{(1+t)^3} dt$$
$$= \frac{-2e^t}{(1+t)^2} + c$$
$$= -2 \cos^4 x e^{\tan^2 x} + c$$

Q75. If curve $y = k \cos \frac{x}{2}$ divides area bounded by $y = \sin x$, x-axis in between $x = 0$ & $x = \pi$, in two equal parts, then $(k + \sqrt{2})$ is : _____.

Ans. 2

Sol. $y = \sin x$, x-axis, $x = 0, x = \pi$

Intersection point $k \cos \frac{x}{2} = \sin x$

$$k \cos \frac{x}{2} = 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

$$\sin \frac{x}{2} = \frac{k}{2}$$

$$\Rightarrow x$$



$$= 2 \sin^{-1} \left(\frac{k}{2} \right)$$

$$A = \int_0^\pi \sin x dx = 2 \frac{A}{2} = \int_{2 \sin^{-1} \frac{k}{2}}^\pi (\sin x - k \cos \frac{x}{2}) dx = \left[-\cos x - 2k \sin \frac{x}{2} \right]_{2 \sin^{-1} \left(\frac{k}{2} \right)}^\pi$$

$$1 = - \left[(-1 + 2k) - \left(\cos \left(2 \sin^{-1} \left(\frac{k}{2} \right) \right) + 2k \sin \left(\sin^{-1} \left(\frac{k}{2} \right) \right) \right) \right]$$

$$= - \left((2k - 1) - \left(1 - 2 \left(\frac{k}{2} \right)^2 + 2k \left(\frac{k}{2} \right) \right) \right) \quad 1 = 2k + 1 - 1 + \frac{k^2}{2} - k^2 - \frac{k^2}{2} + 2k - 1 = 0$$

$$k^2 - 4k + 2 = 0 \quad k = 2 \pm \sqrt{2} \Rightarrow k = 2 - \sqrt{2} \quad k + \sqrt{2} = 2$$

Q76. Let $\int_0^\infty \frac{t^4 dt}{(1+t^2)^6} = \frac{3\pi}{64k}$ then k is equal to _____.

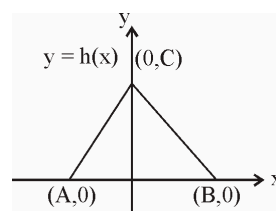
Ans. 8

Sol. Let $I = \int_0^\infty \frac{t^4 dt}{(1+t^2)^6}$

Let $t = \tan \theta$

$$dt = \sec^2 \theta d\theta$$
$$= \int_0^{\pi/2} \frac{\tan^4 \theta \sec^2 \theta d\theta}{\sec^{12} \theta}$$
$$= \int_0^{\pi/2} \sin^4 \theta \cdot \cos^6 \theta d\theta$$
$$\frac{(3 \cdot 1) \cdot (5 \cdot 3 \cdot 1)}{(10 \cdot 8 \cdot 6 \cdot 4 \cdot 2)} \cdot \frac{\pi}{2} = \frac{3\pi}{512}$$
$$\Rightarrow k$$
$$= 8$$

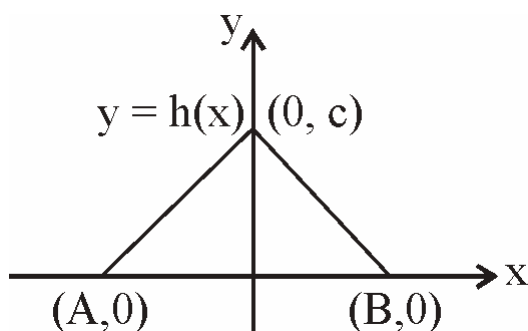
Q77. Let $h(x)$ be the piecewise-linear function graphed in the given figure. If there exist real numbers α , β and γ so that $\int_A^B h(x)f(x)dx = \alpha f(A) + \beta f(B) + \gamma f(0)$ for any twice



continuously differentiable function f , then find the value of $\alpha + \beta + \gamma$.

Ans. 0

Sol.



$$h(x) = \begin{cases} C - \frac{C}{A}x & x < 0 \\ C - \frac{C}{B}x & x \geq 0 \end{cases}$$

$$I = \int_A^B h(x)f''(x)dx = \alpha f(A) + \beta f(B) + \gamma f(0)$$

$$I = \int_A^0 \left(C - \frac{C}{A}x\right)f''(x)dx + \int_0^B \left(C - \frac{C}{B}x\right)f''(x)dx$$

$$= \left[\left(C - \frac{C}{A}x\right)f'(x)\right]_A^0 - \int_A^0 \frac{-C}{A}f'(x)dx + \left[\left(C - \frac{C}{B}x\right)f'(x)\right]_0^B - \int_0^B \frac{-C}{B}f'(x)dx$$

$$Cf'(0) + \frac{C}{A}[f(x)]_A^0 + (-Cf'(0)) + \frac{C}{B}[f(x)]_0^B$$

$$= \frac{C}{A}[f(0) - f(A)] + \frac{C}{B}[f(B) - f(0)]$$

$$= -\frac{C}{A}f(A) + \frac{C}{B}[f(B)] + \left[\frac{C}{A} - \frac{C}{B}\right]f(0)$$

$$\Rightarrow \alpha = -\frac{C}{A}, \beta = \frac{C}{B}, \gamma = \frac{C}{A} - \frac{C}{B}$$

$$\alpha + \beta + \gamma = 0$$

Q78. If the value of the definite integral $\int_0^1 \frac{\tan^{-1} x}{1+x} dx$ is equal to $\frac{\pi \ln 2}{k}$ then find the value of k .

Ans. 8

Sol. Let $I = \int_0^1 \frac{\tan^{-1} x}{1+x} dx$.

Use the substitution $x = \tan t$, so $dx = \sec^2 t dt$ and $\tan^{-1} x = t$.

When $x = 0$, $t = 0$; when $x = 1$, $t = \frac{\pi}{4}$.

$$\text{Thus } I = \int_0^{\pi/4} t \frac{\sec^2 t}{1 + \tan t} dt.$$

Since $\frac{d}{dt} \ln(1 + \tan t) = \frac{\sec^2 t}{1 + \tan t}$, integrate by parts.

Therefore $I = [t \ln(1 + \tan t)]_0^{\pi/4} - \int_0^{\pi/4} \ln(1 + \tan t) dt$.

The boundary term is $\frac{\pi}{4} \ln 2$.

Let $J = \int_0^{\pi/4} \ln(1 + \tan t) dt$.

Using $t \rightarrow \frac{\pi}{4} - t$, we have $1 + \tan(\frac{\pi}{4} - t) = \frac{2}{1 + \tan t}$.

Hence $\ln(1 + \tan(\frac{\pi}{4} - t)) = \ln 2 - \ln(1 + \tan t)$.

This gives $J = \int_0^{\pi/4} (\ln 2 - \ln(1 + \tan t)) dt = \frac{\pi}{4} \ln 2 - J$.

So $2J = \frac{\pi}{4} \ln 2$, hence $J = \frac{\pi}{8} \ln 2$.

Therefore $I = \frac{\pi}{4} \ln 2 - \frac{\pi}{8} \ln 2 = \frac{\pi}{8} \ln 2$.

Comparing with $\frac{\pi \ln 2}{k}$, we get $k = 8$.

Q79. Let e be the eccentricity of hyperbola and $f(e)$ be the eccentricity of its conjugate hyperbola then find the value of $2 \left\{ \int_1^{\sqrt{2}} (f(e) + f(f(e))) de \right\}$ _____.

Ans. 3

Sol. -

Q80. If $I_1 = \int_0^{100\pi} (\sin(\cos x) + \cos(\sin x)) dx$ & $I_2 = \int_0^1 \frac{\cos x dx}{\sqrt{1-x^2}}$, then $\frac{I_1}{50I_2}$ is equal to _____.

Ans. 4

Sol. $f(x) = \sin(\cos x) + \cos(\sin x)$ is periodic with period 2π

$\Rightarrow$ From Jack property

$$I_1 = 50 \int_0^{2\pi} (\sin(\cos x) + \cos(\sin x)) dx$$

$$I_1 = 100 \int_0^{\pi} (\sin(\cos x) + \cos(\sin x)) dx \quad \dots (i)$$

(From queen property $\therefore f(x) = f(2\pi - x)$)

$$I_1 = 100 \int_0^{\pi} (-\sin(\cos x) + \cos(\sin x)) dx \quad \dots (ii)$$

(From king property)

Add (i) & (ii)

$$2I_1 = 200 \int_0^{\pi} \cos(\sin x) dx$$

$$I_1 = 100 \int_0^{\pi} \cos(\sin x) dx$$

$$I_1 = 200 \int_0^{\pi/2} \cos(\sin x) dx$$

(Queen property)

Put $\sin x = t$

$$\begin{aligned}
\cos x dx &= dt \\
dx &= \frac{dt}{\sqrt{1-t^2}} \\
&\Rightarrow I_1 \\
&= 200 \int_0^1 \frac{\cos t dt}{\sqrt{1-t^2}} \\
\Rightarrow I_1 &= 200 I_2 \\
\Rightarrow \frac{I_1}{50 I_2} &= 4
\end{aligned}$$

Q81. A polynomial function $f: R \rightarrow R$ satisfies $\int_0^1 f(x)(x - f(x))dx = \frac{1}{12}$ then $[f(1) + f(2) + f(3)]$ is equal to _____.

(where $[\cdot]$ denotes the greatest integer function)

Ans. 3

Sol. Given $\int_0^1 f(x)(x - f(x)) dx = \frac{1}{12}$.

Rewrite the integrand by completing the square:

$$\begin{aligned}
f(x)(x - f(x)) &= f(x)x - f(x)^2 \\
&= \frac{x^2}{4} - \left(f(x) - \frac{x}{2}\right)^2.
\end{aligned}$$

Therefore,

$$\int_0^1 f(x)(x - f(x)) dx = \int_0^1 \frac{x^2}{4} dx - \int_0^1 \left(f(x) - \frac{x}{2}\right)^2 dx.$$

$$\text{Now } \int_0^1 \frac{x^2}{4} dx = \frac{1}{4} \cdot \frac{1}{3} = \frac{1}{12}.$$

So the given equation becomes

$$\frac{1}{12} - \int_0^1 \left(f(x) - \frac{x}{2}\right)^2 dx = \frac{1}{12}.$$

$$\text{Hence } \int_0^1 \left(f(x) - \frac{x}{2}\right)^2 dx = 0.$$

Since the integrand is a square of a continuous function, this implies

$$f(x) - \frac{x}{2} = 0 \text{ for all } x \in [0, 1].$$

As f is a polynomial, it must be identically

$$f(x) = \frac{x}{2} \text{ for all real } x.$$

Thus,

$$f(1) + f(2) + f(3) = \frac{1}{2} + 1 + \frac{3}{2} = 3.$$

Therefore,

$$[f(1) + f(2) + f(3)] = [3] = 3.$$

Q82. If $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \cdot \cos(nx) dx$, then $\left(\frac{I_n}{I_{n+1}}\right)^3$ is equal to _____.

Ans. 8

Sol.

$$I_{n+1} = \int_0^{\frac{\pi}{2}} \cos^{n+1} x \cdot \cos(nx + x) dx$$

$$I_{n+1} = \int_0^{\frac{\pi}{2}} \cos^{n+1} x \cdot (\cos nx \cos x - \sin nx \sin x) dx$$

$$I_{n+1} = \int_0^{\frac{\pi}{2}} [\cos^n x \cdot \cos nx (1 - \sin^2 x) - \cos^{n+1} x \sin nx \cdot \sin x] dx$$

$$I_{n+1} = \int_0^{\frac{\pi}{2}} \cos^n x \cos nx dx - \int_0^{\frac{\pi}{2}} [\cos^n x \cdot \cos nx \cdot \sin^2 x + \cos^{n+1} x \sin nx \cdot \sin x] dx$$

$$I_{n+1} = I_n - \int_0^{\frac{\pi}{2}} \underbrace{\cos^n x \cdot \sin x}_{II} \cdot \underbrace{\sin(n+1)x}_I dx$$

$$I_{n+1} = I_n - \left[\sin(n+1)x \left(-\frac{\cos^{n+1} x}{n+1} \right) \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} (n+1) \cos(n+1)x \frac{\cos^{n+1} x}{(n+1)} dx$$

$$I_{n+1} = I_n - I_{n+1}$$

$$\Rightarrow 2I_{n+1}$$

$$= I_n$$

$$\Rightarrow \frac{I_n}{I_{n+1}}$$

$$= 2$$

Q83. If $f(x) = \frac{x}{x-1}$, then $G(x) = \int \underbrace{\text{fofof} \dots \dots (x)}_{2017 \text{ times}} dx$, where $G(2) = 1$, then the value of $[G(e+1)]$ is _____.

(Where $[.]$ denotes greatest integer function)

Ans. 3

Sol.

$$G(x) = \int \underbrace{\text{fofof} \dots \dots (x)}_{2017 \text{ times}} dx$$

$$= \int \frac{x}{x-1} dx$$

$$G(x) = \int \left(\frac{x-1}{x-1} + \frac{1}{x-1} \right) dx = x + \ln|x-1| + C$$

$$G(2) = 2 + 0 + C \Rightarrow C = -1$$

$$G(x) = (x-1) + \ln|x-1|$$

$$[G(e+1)] = [e + \ln e] = [e + 1] = 3$$

Q84. Area of the region in $x-y$ plane such that $\sin^2 x - \sin x \sin y + \sin^2 y \leq \frac{3}{4}$, where $x, y \in \left[0, \frac{\pi}{2}\right]$, is $\frac{\pi^2}{k}$, then the value of k is.....

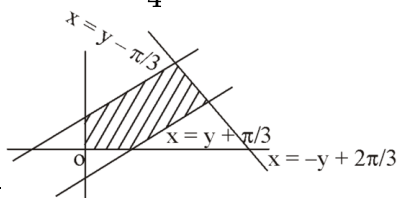
Ans. 6

Sol. If $\sin^2 x - \sin x \sin y + \sin^2 y = \frac{3}{4}$

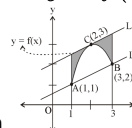
$$\Rightarrow \sin x$$

$$= \frac{\sin y}{2} \pm \frac{\sqrt{3} \cos y}{2}$$

$$= \sin \left(y \pm \frac{\pi}{3} \right)$$



Q85. The following figure shows the graph of a continuous function $y = f(x)$ on the interval $[1, 3]$. The points A, B, C have co-ordinates $(1, 1)$, $(3, 2)$, $(2, 3)$ respectively and the lines L_1 and L_2 are parallel with L_1 being tangents to the curve at C. If the area under the graph of $y = f(x)$ from $x = 1$ to $x = 3$



is 4 square units, then find the area (in square units) of shaded region.

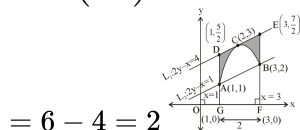
Ans. 2

Sol. Area of shaded region

$$= (\text{area of trapezium DEFG}) - (\text{area under } f(x))$$

$$= \frac{1}{2} \left(\frac{5}{2} + \frac{7}{2} \right) \times 2 - 4$$

$$= \frac{1}{2} \left(\frac{12}{2} \right) \times 2 - 4$$



$$= 6 - 4 = 2$$

Q86. $\int_{-1}^1 \frac{(3x^2 + 5x^4)dx}{1 + (1 + x^2 + x^4)x}$ is equal to _____.

Ans. 4

Sol.

$$I = \int_{-1}^1 \frac{(3x^2 + 5x^4)dx}{1 + (1 + x^2 + x^4)x}$$

$$I = \int_{-1}^1 \frac{(3x^2 + 5x^4)[1 + x^2 + x^4]dx}{1 + (1 + x^2 + x^4)x} \quad (\text{using } \int_a^b f(x)dx = \int_a^b f(a + b - x)dx)$$

$$2I = \int_{-1}^1 (3x^2 + 5x^4)dx = x^3 + x^5 \Big|_{-1}^1 = 4$$

Q87. If the differential equation of a curve passing through $\left(0, -\frac{\pi}{4}\right)$ and $(t, 0)$ is

$$\cos y \left(\frac{dy}{dx} + e^{-x} \right) + \sin y \left(e^{-x} - \frac{dy}{dx} \right) = e^{-x}, \text{ then the value of } te^{e^{-t}} \text{ is } \underline{\hspace{2cm}}.$$

Ans. 1

Sol. $(\cos y - \sin y)dy + (\cos y + \sin y)e^{-x}dx = e^{-x} \cdot dx$

$$\text{Let } \cos y + \sin y = t$$

$$(\cos y - \sin y)dy = dt$$

$$\frac{dt}{dx} + te^{-x} = e^{-x}$$

$$\begin{aligned}\text{I.F.} &= e^{e^{-x}} \\ \Rightarrow te^{e^{-x}} &= x\end{aligned}$$

Q88. If $P = \int e^{3x} \cos(2013x) dx$ and $Q = \int e^{3x} \sin(2013x) dx$ (where P & Q both have integration constant zero) & $f(x) = \ln(P^2 + Q^2)(3^2 + 2013^2)$ then $f(1)$ is _____.

Ans. 6

Sol. $P = \frac{e^{3x}}{3^2 + 2013^2} (3 \cos 2013x + 2013 \sin 2013x)$

$$Q = \frac{e^{3x}}{3^2 + 2013^2} (3 \sin 2013x - 2013 \cos 2013x)$$

$$P^2 + Q^2 = \frac{e^{6x}}{(3^2 + 2013^2)^2} (3^2 + 2013^2)$$

$$P^2 + Q^2 = \frac{e^{6x}}{(3^2 + 2013^2)}$$

$$\ln((P^2 + Q^2)(3^2 + 2013^2)) = 6x$$

$$f(1) = 6$$

Q89. Water is flowing in a container at a rate proportional to the square of the amount of water present inside the container. If at $t = 0$, volume of water inside the tank is V_0 & it takes 6 hours for the volume to become $2V_0$, then the time (in hours) taken for the volume to become $4V_0$ is equal to _____.

Ans. 9

Sol. Let $\frac{dV}{dt} = kV^2$

$$\Rightarrow \int \frac{dV}{V^2} = \int k dt \Rightarrow -\frac{1}{V} = kt + C$$

$$\text{at } t = 0, V = V_0$$

$$\Rightarrow C = -\frac{1}{V_0} \Rightarrow \frac{1}{V_0} - \frac{1}{V} = kt$$

$$\text{at } t = 6, V = 2V_0$$

$$\Rightarrow 6k = \frac{1}{2V_0} \Rightarrow k = \frac{1}{12V_0}$$

$$\text{when } V = 4V_0$$

$$\Rightarrow kt = \frac{3}{4V_0} = \frac{3 \times 12V_0}{4V_0 \times 1}$$

$$t = 9 \text{ hours}$$

Q90. Let $f : [0, 2\pi] \rightarrow R$ be given as $f(x) = \sin x \forall x \in \left[0, \frac{\pi}{2}\right]$, $f(x) + f(\pi - x) = 2 \forall x \in \left(\frac{\pi}{2}, \pi\right]$ and $f(x) = f(2\pi - x) \forall x \in (\pi, 2\pi)$, then area bounded by curve $y = f(x)$ and x-axis is $k\pi$ square units where k is :

Ans. 2

Sol.

$$f(x) = \begin{cases} \sin x & x \in [0, \frac{\pi}{2}] \\ 2 - \sin x & x \in (\frac{\pi}{2}, \pi] \\ 2 + \sin x & x \in (\pi, \frac{3\pi}{2}] \\ -\sin x & x \in (\frac{3\pi}{2}, 2\pi] \end{cases}$$

Area

$$= \int_0^{\pi/2} \sin x dx + \int_{\pi/2}^{\pi} (2 - \sin x) dx + \int_{\pi}^{3\pi/2} (2 + \sin x) dx + \int_{3\pi/2}^{2\pi} (-\sin x) dx = 1 + \pi - 1 + \pi - 1 + 1 = 2\pi \text{ sq. units.}$$

Q91. Let f be a differentiable function satisfying $f'(x) = f^2(x)$ and $f(0) = -1$.

If $\sqrt{\frac{2}{\pi}} \int_0^1 \left(\frac{(1-x)}{f(x)(x^4+1)} + \frac{2}{x^4+1} \right) dx = \frac{p}{q}$ where $p, q \in N$ then find least value of $(p+q)$

_____.

Ans. 3

Sol. $f'(x) = f^2(x)$

$$\Rightarrow \frac{f'(x)}{f^2(x)}$$

$$= 1$$

$$\Rightarrow \int \frac{f'(x)}{f^2(x)} dx$$

$$\int 1 \cdot dx + C \Rightarrow -\frac{1}{f(x)} = x + C$$

$$\text{at } x = 0, f(0) = -1 \Rightarrow C = 1$$

$$\therefore f(x) = -\frac{1}{x+1}$$

$$I = \sqrt{\frac{2}{\pi}} \int_0^1 \left[\frac{1-x}{-\frac{1}{x+1}(x^4+1)} + \frac{2}{x^4+1} \right] dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^1 \left(\frac{x^2-1}{x^4+1} + \frac{2}{x^4+1} \right) dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^1 \frac{x^2+1}{x^4+1} dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^1 \frac{1 + \frac{1}{x^2}}{x^2 + \frac{1}{x^2}} dx$$

$$= \sqrt{\frac{2}{\pi}} \int_0^1 \frac{(1 + \frac{1}{x^2})}{(x - \frac{1}{x})^2 + 2} dx$$

$$x - \frac{1}{x} = t \Rightarrow \left(1 + \frac{1}{x^2} \right) dx = dt$$

$$= \sqrt{\frac{2}{\pi}} \int_{-\infty}^0 \frac{at}{t^2+2}$$

$$= \sqrt{\frac{2}{\pi}} \left[\frac{1}{\sqrt{2}} \tan^{-1} \left(\frac{t}{\sqrt{2}} \right) \right]_{-\infty}^0$$

$$= \frac{1}{\pi} \left(0 - \left(-\frac{\pi}{2} \right) \right)$$

$$= \frac{1}{\pi} \cdot \frac{\pi}{2}$$

$$= \frac{1}{2}$$

$$\therefore (p+q) = 3$$

Q92. Let $y^2(x)g(x)g'(x) + y(x)y'(x) = g(x)g'(x)$, $y(0) = 0$, $x \in \mathbb{R}$ where $f'(x)$ denotes $\frac{d}{dx}f(x)$ and $g(x)$ is a given non-constant differential function on $\mathbb{R}$ with $g(0) = g(e) = 1$. Then the value of $y(e)$ is _____.

Ans. 0

Sol. Let $y^2(x)g(x)g'(x) + y(x)y'(x) = g(x)g'(x) \Rightarrow$ multiply by $2e^{g^2(x)}$
 $\Rightarrow 2y^2(x)g(x)g'(x)e^{g^2(x)} + 2y(x)y'(x)e^{g^2(x)}$
 $= 2g(x)g'(x)e^{g^2(x)}$
 $\Rightarrow \frac{d}{dx} [e^{g^2(x)}y^2(x)]$
 $= 2g(x)g'(x)e^{g^2(x)}$
 $\Rightarrow e^{g^2(x)}y^2(x)$
 $= \int 2g(x)g'(x)e^{g^2(x)}dx$
 $g^2(x) = t$
 $2g(x)g'(x)dx = dt$
 $e^{g^2(x)}y^2(x) = \int e^t dt$
 $e^{g^2(x)}y^2(x) = e^{g^2(x)} + c$
 $x = 0, e^{g^2(0)}y^2(0) = e^{g^2(0)} + c, g(0) = 1$
 $e^1(0) = e^1 + c \therefore c = -e$
 Put $x = e$
 $e^{g^2(e)}y^2(e) = e^{g^2(e)} - e, g(e) = 1$
 $ey^2(e) = e - e, y(e) = 0$

Q93. Let $y = f(x)$ be a curve passing through (e, e^e) which satisfy the differential equation $(2ny + xy \log_e x)dx - x \log_e x dy = 0, \log_e x dy = 0, x > 0, y > 0$. If $g(x) = \lim_{n \rightarrow \infty} f(x)$, then

$$\int_{\frac{1}{e}}^e g(x)dx =$$

Ans. 0

Sol. $(2ny + xy \log_e x)dx - x \log_e x dy = 0$
 $\Rightarrow \frac{dy}{y}$
 $= \left(\frac{2n}{x \log_e x} + \frac{1}{x} \right) dx$
 $\Rightarrow \log y = 2n \log |\log x| + \log x + c$
 $\Rightarrow \log y = 2n \log |\log x| + \log x + c$
 $\Rightarrow y = e^{2n \log(\log x) + \log x + c}$
 $\Rightarrow y = e^{\log(\log x)^{2n}} \cdot e^{\log x} \cdot e^c$
 $\Rightarrow y = (\log x)^{2n} \cdot x \cdot A$
 Curve passes through (e, e^e)
 $e^e = (\log e)^{2n} \cdot e \cdot A$
 $e^e = 1^{2n} \cdot e \cdot A$
 $e^e = e \cdot A$
 $A = e^{e-1}$
 So, $f(x) = xe^{e-1}(\log x)^{2n}$
 Now, $g(x) = \lim_{n \rightarrow \infty} f(x) = \lim_{n \rightarrow \infty} xe^{e-1}(\log x)^{2n}$

$$g(x) = \begin{cases} \infty & \text{if } x < \frac{1}{e} \\ 0 & \text{if } \frac{1}{e} < x < e \\ \infty & \text{if } x > e \end{cases}$$

$$\text{Therefore, } \int_{\frac{1}{e}}^e g(x) dx = 0$$

- Q94. Let two curves $y = f(x)$ with $f(0) = 1$ and $g(x) = \int_{-\infty}^x f(t) dt$ with $g(0) = 4$ are such that, their tangents at the points with equal abscissa intersects on x-axis, then area bounded by $y = f(x)$, $y = g(x)$, $x = 1$ and y axis is $k(e^{1/4} - 1)$ sq. unit. Tangents are drawn to the parabola $y^2 = kx$ at the points A, B and C such that the three tangents form a triangle PQR. If θ_1, θ_2 and θ_3 be the inclinations of these tangents with the axis of x such that their cotangents form an arithmetical progression (in the same order) with common difference 2. Find the area of the triangle PQR

_____ . x

Ans. -

Sol. -

- Q95. Solve the differential equation $\frac{dy}{dx} = \frac{(x+y)^2}{(x+2)(y-2)}$

Ans. -

Sol. Ans. $X^4 \left(1 + \frac{2Y}{X}\right) = Ke^{2Y/X}$ where $X = x + 2$, and $Y = y - 2$

$$\text{Sol. } \frac{dy}{dx} = \frac{(x+y)^2}{(x+2)(y-2)}$$

$$= \frac{[(x+2) + (y-2)]^2}{(x+2)(y-2)}$$

$$\frac{dy}{dx} = \left(\frac{x+2}{y-2}\right) + \left(\frac{y-2}{x+2}\right) + 2$$

Let $x = X - 2, y = Y + 2$

$$\frac{dY}{dX} = \frac{X}{Y} + \frac{Y}{X} + 2$$

Suppose $Y = vX$

$$\frac{dY}{dX} = v + \frac{Xdv}{dX}$$

$$v + X \frac{dv}{dX} = \frac{1}{v} + v + 2$$

$$X \frac{dv}{dX} = \frac{1+2v}{v}$$

$$\left(\frac{v}{1+2v}\right) dv = \frac{dX}{X}$$

$$\frac{1}{2} \left[\frac{2v+1-1}{2v+1} \right] dv = \frac{dX}{X}$$

$$\left[1 - \frac{1}{2v+1} \right] dv = 2 \frac{dX}{X}$$

$$v - \frac{1}{2} \ln |2v+1| = 2 \ln |X| + c$$

$$\frac{Y}{X} - \frac{1}{2} \ln \left| \frac{2Y}{X} + 1 \right| = 2 \ln |X| + c$$

$$d + 2 \frac{Y}{X} = \ln \left| \frac{2Y + X}{X} \right| + \ln X^4$$

$$ke^{\frac{2Y}{X}} = X^4 \left(1 + \frac{2Y}{X} \right) \text{ (where k is constant)}$$

Q96. A curve $y = f(x)$ is such that $f(x) \geq 0$ and $f(0) = 0$ and bonds a curve linear trapezoid with the base $[0, x]$ whose area is proportional to $(n + 1)^{\text{th}}$ power of $f(x)$. If $f(1) = 1$, then $\{f(10)\}^n$ is k , then the value of $k/4$ is

Ans. -

Sol. Area of curve linear trapezoid OABC = $\int_0^x f(x) dx$

$$\text{According to } = \int_0^x f(x) dx = k \{f(x)\}^{n+1}$$

Where k is constant of proportionality

Differentiating both sides w.r.t. x

$$f(x) = k(n + 1)[f(x)]^n f'(x)$$

$$\text{Or } \{f(x)\}^{n-1} f'(x) = \frac{1}{k(n + 1)}$$

Integrating both sides w.r.t. x

$$\frac{\{f(x)\}^n}{n} = \frac{x}{k(n + 1)} + c$$

$$\text{Putting } x = 0, \frac{\{f(0)\}^n}{n} = 0 + c$$

$$\Rightarrow 0 = 0 + c \quad [\because f(0) = 0]$$

$$\therefore \frac{\{f(x)\}^n}{n} = \frac{x}{k(n + 1)}$$

$$\Rightarrow \{f(x)\}^n = \frac{nx}{k(n + 1)}$$

$$= \frac{nx}{k(n + 1)}$$

$$\text{Again, putting } x = 1, \text{ then } \{f(1)\}^n = \frac{n}{k(n + 1)} = 0$$

$$\text{From equation (1) } \{f(x)\}^n = x, \{f(10)\}^n = 10$$



Q97. Consider the function $f(x) = x^{\sin(x) + \cos(x)} \forall x > 0$. Prove that $0.4 \leq \int_0^1 f(x) dx \leq 0.5$.

Ans. -

Sol. It is easy to see that for $0 \leq x \leq 1$, we have $1 \leq \cos(x) + \sin(x) \leq \sqrt{2}$, and

$$x^1 \geq x^{\cos(x) + \sin(x)}$$

so $\geq x^{\sqrt{2}}$ *As all three functions are non-negative in $[0, 1]$* , we can integrate the

inequalities over that interval to get

$$\frac{1}{2} \geq \int_0^1 f(x) dx$$

$$\geq \frac{1}{\sqrt{2} + 1} > \frac{1}{1.5 + 1}$$

$$= 0.4$$

Q98. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a continuous function such that for all $x \in (0, \infty)$, $f(2x) = f(x)$. Show that

the function g defined by the equation $g(x) = \int_x^{2x} \frac{f(t)}{t} dt$ for $x > 0$ is a constant function.

Ans. -

Sol. The function $t \rightarrow \frac{f(t)}{t}$ is continuous on $(0, \infty)$. So, g is continuously differentiable on $(0, \infty)$. We calculate the derivative of g :

$$\begin{aligned} g'(x) &= \frac{d}{dx} \left(\int_x^{2x} \frac{f(t)}{t} dt \right) \\ &= 2 \frac{f(2x)}{2x} - \frac{f(x)}{x} \\ &= \frac{f(2x) - f(x)}{x} \\ &= 0 \end{aligned}$$

So, g is a constant function.

Q99. Let $f(x)$ be a function defined by $f(x) = \int_3^x (t^2 - 7t + 12)dt$, $3 \leq x \leq 6$. If the range of $f(x)$ is $[a, b]$, then the value of $\left(3a + \frac{b}{9}\right)$ is _____.

Ans. -

Sol. -

Q100. Match the following column :

Column-I		Column-II	
(a) The function $f(x) = \frac{e^{x \cos x} - 1 - x}{\sin x^2}$ is not defined at $x = 0$ The value of $f(0)$ so that f is continuous at $x = 0$ is	(p)	-1	
(b) The value of the definite integral $\int_0^1 \frac{dx}{\sqrt{x} + \sqrt[3]{x}}$ equals $a + b \ln 2$ where a and b are integers then $(a + b)$ equals	(q)	0	
(c) Given $e^n \int_0^n \frac{\sec^2 \theta - \tan \theta}{e^\theta} d\theta = 1$ then the value of $\tan(n)$ is equal to	(r)	$\frac{1}{2}$	
(d) Let $a_n = \int_{1/n+1}^{1/n} \tan^{-1}(nx) dx$ and $b_n = \int_{1/n+1}^{1/n} \sin^{-1}(nx) dx$ then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$ has the value equal to	(s)	1	

Choose the correct option :

- (A) (a)-(p), (b)-(q), (c)-(r), (d)-(s) (B) (a)-(q), (b)-(p), (c)-(r), (d)-(s)
 (C) (a)-(q), (b)-(p), (c)-(s), (d)-(r) (D) (a)-(s), (b)-(q), (c)-(r), (d)-(p)

Ans. A

Sol. (A) $f(x) = \frac{e^{x \cos x} - 1 - x}{\sin^2 x}, x \neq 0$

$f(x)$ is continuous at $x = 0$

if LHS = RHS = $f(0)$

$$\begin{aligned} f(0) &= \lim_{x \rightarrow 0} \frac{(e^{x \cos x} - 1) - x}{\sin^2 x} \\ &= \lim_{x \rightarrow 0} \frac{\left(\left(1 + (x \cos x) + \frac{(x \cos x)^2}{2} + \dots \right) - 1 \right) - x}{\sin^2 x} \\ &= \lim_{x \rightarrow 0} \frac{x(\cos x - 1) + \frac{(x \cos x)^2}{2} + \dots}{\sin^2 x} \\ &= \lim_{x \rightarrow 0} \frac{x \left(-2 \sin^2 \frac{x}{2} \right) + \frac{(x \cos x)^2}{2!} + \dots}{\sin^2 x} \\ &= \lim_{x \rightarrow 0} \left(\frac{-2x \sin^2 \frac{x}{2}}{\sin^2 x} + \frac{x^2 \cos^2 x}{2 \sin^2 x} \right) \\ \lim_{x \rightarrow 0} \left(0 + \frac{1}{2} \right) &= \frac{1}{2} \end{aligned}$$

(B) $I = \int_0^1 \frac{dx}{\sqrt{x} + 3\sqrt[3]{x}} = a + b \ln^2$

put $x = t^6 \Rightarrow dx = 6t^5 dt$

$$I = \int_0^1 \frac{6t^5 dt}{t^3 + t^2} = 6 \int_0^1 \frac{t^3}{t+1} dt$$

$$\begin{aligned} I &= 6 \int_0^1 \frac{t^3 + 1 - 1}{t+1} dt \\ &= 6 \int_0^1 \left((t^2 - t + 1) - \left(\frac{1}{t+1} \right) \right) dt \\ &= 6 \left[\frac{t^3}{3} - \frac{t^2}{2} + t - \ln^{(t+1)} \right]_0^1 \\ &= 6 \left[\frac{1}{3} - \frac{1}{2} + 1 - \ln^2 \right] \\ &= 6 \left[\frac{2 - 3 + 6 - 6 \ln^2}{6} \right] \end{aligned}$$

$$I = 5 - \ln^2$$

$$a = 5, b = -6$$

$$(a + b) = -1$$

(c) $I = e^n \int_0^n \frac{\sec^2 \theta - \tan \theta}{e^\theta} d\theta = 1$

$$I = e^n \int_0^n e^{-\theta} (\sec^2 \theta - \tan \theta) d\theta$$

put $\theta = -t$

$$I = -e^n \int_0^{-n} e^t [\sec^2 t + \tan t] dt$$

$$= -e^n [e^t \tan t]_0^{-n}$$

$$I = -e^n [e^{-n} \tan(-n)]$$

$$I = \tan(n)$$

$$\Rightarrow \tan(n) = 1$$

$$(D) a_n = \int_{\frac{1}{n+1}}^{\frac{1}{n}} \tan^{-1}(nx) dx, b_n = \int_{\frac{1}{n+1}}^{\frac{1}{n}} \sin^{-1}(n_x) dx$$

$$= 6 \int_0^1 \left(t^2 - t + 1 - \frac{1}{t+1} \right) dt$$

$$= 6 \left[\frac{t^3}{3} - \frac{t^2}{2} + t - \ln(t+1) \right]_0^1$$

$$= 6 \left[\frac{1}{3} - \frac{1}{2} + 1 - \ln 2 \right]$$

$$= 6 \left[\frac{2-3+6}{6} - \ln 2 \right]$$

$$= 5 - 6 \ln 2$$

$$\text{So } a = 5, b = -6$$

$$(C) \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\int_{1/(n+1)}^{1/n} \tan^{-1}(nx) dx}{\int_{1/(n+1)}^{1/n} \sin^{-1}(nx) dx}$$

$$= \lim_{n \rightarrow \infty} \frac{\tan^{-1} \left(n \cdot \frac{1}{n} \right) \cdot \left(-\frac{1}{n^2} \right) - \tan^{-1} \left(n \cdot \frac{1}{n+1} \right) \cdot \left(-\frac{1}{(n+1)^2} \right)}{\sin^{-1} \left(n \cdot \frac{1}{n} \right) \cdot \left(-\frac{1}{n^2} \right) - \sin^{-1} \left(n \cdot \frac{1}{n+1} \right) \cdot \left(-\frac{1}{(n+1)^2} \right)}$$

(using lebnitz rule)

$$= \lim_{n \rightarrow \infty} \frac{\frac{\pi}{4} \times \frac{1}{n^2} - \tan^{-1} \left(\frac{n}{n+1} \right) \times \frac{1}{(n+1)^2}}{\frac{\pi}{2} \times \frac{1}{n^2} - \sin^{-1} \left(\frac{n}{n+1} \right) \times \frac{1}{(n+1)^2}}$$

$$= \lim_{n \rightarrow \infty} \frac{\pi(n+1)^2 - 4n^2 \tan^{-1} \left(\frac{n}{n+1} \right)}{2\pi(n+1)^2 - 2n^2 \sin^{-1} \left(\frac{n}{n+1} \right)}$$

$$= \frac{1}{2}$$

Q101. Match the following column :

Column-I Column-II

[I02/5/171.] Match the following column :

Column-I	Column-II
(a) Value of $\int_0^1 \frac{dx}{\sqrt{1-x^2}(1+\sqrt{1-x^2})}$ is equal to	(p) 2
(b) $f: \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x^5 + e^2 + e^{x^3}$ & $g(x) = f^{-1}(x)$, then value of $g'(2)$ is equal to	(q) 4
(c) $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \left(1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{2\sqrt{n}} \right)$ equal to	(r) 12
(d) If $f(x) = \int_{x-1}^{2x} (t^3 - 2t + 4) dt$, then value of $f'(1)$ is equal to	(s) 1

Choose the correct option

(A) a - s; b - p; c - q; d - r (B) a - p; b - q; c - s; d - r (C) a - q; b - p; c - r; d - s

(D) p - r; b - s; c - q; d - p

Ans. C

Sol. $g'(f(x)) = \frac{1}{f'(x)}$

& $f(0) = 2$

$\therefore g'(2) = 2$

(R) $T_r = \frac{1}{\sqrt{n}\sqrt{r}} = \frac{1}{n(\sqrt{r}/n)}$

$\therefore \text{Sum} = \lim_{n \rightarrow \infty} \sum_{r=1}^{4n} \frac{1}{n\sqrt{r}/n} = \int_0^4 \frac{1}{\sqrt{x}} dx$
 $= 4$

(S) $f'(x) = [(2x)^3 - 2(2x) + 4] \times 2$
 $- [(x-1)^3 - 2(x-1) + 4]$

$f'(1) = 12$

Q102. Match the following column :

Column-I

Column-II

(a) Let $\int \cos^5 x dx = \sin x + \frac{\sin^3 x}{m} + \frac{\sin^5 x}{n} + c$, then $n + 2m$ is equal to

(p) 2

(b) Let $\int \frac{xdx}{x^2 - 8x + 15} = \ln \left| \frac{(x-5)^a}{(x-3)^b} \right| + c$, then $a + b$ is equal to

(q) 3

(c) Let $y = f(t)$ & $x = g(t)$ where t is a parameter & f, g are twice derivable functions. If $f'(1) = 0$, $g'(1) = 1$ & $f''(1) = 2$ then, $\frac{d^2y}{dx^2}$ at $t = 1$ is equal to

(r) 0

(d) Let $L = \lim_{x \rightarrow 0^+} (-\ln x)^{nx}$ then $\ln L$ is equal to

(s) -2

(t) Natural number

(A) (a) - pt; (b) -t; (c) - pt; (d) - r (B) (a) - pt; (b) -t; (c) - pt; (d) - t

(C) (a) - st; (b) -r; (c) - qt; (d) - qrs (D) (a) - pt; (b) -qs; (c) - rt; (d) - rs

Ans. A

Sol. $\int \cos^5 x dx = \int (1 - \sin^2 x)^2 \cos x dx$

Let $\sin x = t$

$\Rightarrow \cos x dx = dt$

$= \int (1 - t^2)^2 dt = \int (1 + t^4 - 2t^2) dt$

$= t - \frac{2t^3}{3} + \frac{t^5}{5} + c$

$= \sin x - \frac{2}{3} \sin^3 x + \frac{\sin^5 x}{5} + c$

$\Rightarrow n + 2m = 5 - \frac{3}{2} \times 2 = 2$

$$I = \int \frac{x dx}{(x-3)(x-5)}$$

$$\Rightarrow 2I = \int \frac{(5(x-3) - 3(x-5)) dx}{(x-3)(x-5)}$$

$$= \ln|x-5|^5 - \ln|x-3|^3$$

$$I = \ln \left| \frac{(x-5)^{5/2}}{(x-3)^{3/2}} \right| + c \Rightarrow a + b = 4$$

$$\frac{dy}{dx} = \frac{f'(t)}{g'(t)}$$

$$\frac{d^2y}{dx^2} = \frac{(g'(t) \cdot f''(t) - f'(t) \cdot g''(t))}{(g'(t))^2} \cdot \frac{1}{g'(t)}$$

Put $t = 1$

$$\frac{d^2y}{dx^2} = \frac{2}{1} = 2$$

$$(D) L = \lim_{x \rightarrow 0^+} (-\ln x)^{\frac{1}{\ln x}}$$

$$e^{\lim_{x \rightarrow 0^+} \frac{\ln(-\ln x)}{\ln x}}$$

Applying L'Hospital rule

$$L = e^{\lim_{x \rightarrow 0^+} \frac{\frac{1}{-\ln x} \cdot \frac{-1}{x}}{\frac{1}{x}}}$$

$$= e^0$$

$$= 1$$

$$\Rightarrow \ln L = 0$$

Q103. Match the following column :

Column-I

Column-II

(a) Let $\int \cos^5 x dx = \sin x + \frac{\sin^3 x}{m} + \frac{\sin^5 x}{n} + c$, then $n + 2m$ is equal to

(p) 2

(b) Let $\int \frac{x dx}{x^2 - 8x + 15} = \ln \left| \frac{(x-5)^a}{(x-3)^b} \right| + c$, then $a + b$ is equal to

(q) 3

(c) Let $y = f(t)$ & $x = g(t)$ where t is a parameter & f, g are twice derivable functions. If $f'(1) = 0, g'(1) = 1$ & $f''(1) = 2$ then, $\frac{d^2y}{dx^2}$ at $t = 1$ is equal to

(r) 0

(d) Let $L = \lim_{x \rightarrow 0^-} (-\ln x)^{\frac{1}{\ln x}}$ then $\ln L$ is equal to

(s) -2

(t) Natural number

(A) $a \rightarrow p; b \rightarrow r; c \rightarrow q; d \rightarrow s$ (B) $a \rightarrow s; b \rightarrow q; c \rightarrow p; d \rightarrow r$

(C) $a \rightarrow q; b \rightarrow p; c \rightarrow s; d \rightarrow t$ (D) $a \rightarrow t; b \rightarrow s; c \rightarrow r; d \rightarrow p$

Ans. B

Sol. (A) $\int \frac{dx}{(x^2 + 1)\sqrt{x^2 + 2}} = \int \frac{1/x^3 dx}{(1 + \frac{1}{x^2})\sqrt{1 + 2/x^2}}$

Let $1 + 2/x^2 = t^2$

$$\begin{aligned} I &= \int \frac{1}{(x+2)\sqrt{x^2+6x+7}} dx \\ \text{(B)} \quad &= \int \frac{dx}{(x+2)\sqrt{(x+2)^2+2(x+2)-1}} \end{aligned}$$

$$I = \sin^{-1} \left(\frac{x+1}{\sqrt{2}(x+2)} \right) + C$$

$$\text{(C)} \quad \int \frac{x^4+x^8}{(1-x^4)^{7/2}} = \int \frac{(x+\frac{1}{x^3})dx}{(\frac{1}{x^2}-x^2)^{7/2}}$$

$$\text{Let } \frac{1}{x^2} - x^2 = t^2$$

$$\text{(D)} \quad \text{Let } \sqrt{x} = \cos 2\theta \Rightarrow dx = -2 \sin 4\theta d\theta$$

$$\int \sqrt{\frac{1-\sqrt{x}}{1+\sqrt{x}}} dx = -2 \int \tan \theta \sin 4\theta d\theta$$

$$= t - \frac{2t^3}{3} + \frac{t^5}{5} + c$$

$$= \sin x - \frac{2}{3} \sin^3 x + \frac{\sin^5 x}{5} + c$$

$$\Rightarrow n+2m = 5 - \frac{3}{2} \times 2 = 2$$

$$\begin{aligned} \text{(B)} \quad I &= \int \frac{x dx}{(x-3)(x-5)} \\ \Rightarrow 2I &= \int \frac{(5(x-3) - 3(x-5))dx}{(x-3)(x-5)} \end{aligned}$$

$$= \ln |x-5|^5 - \ln |x-3|^3$$

$$I = \ln \left| \frac{(x-5)^{5/2}}{(x-3)^{3/2}} \right| + c$$

$$\Rightarrow a+b=4$$

$$\text{(C)} \quad \frac{dy}{dx} = \frac{f'(t)}{g'(t)}$$

$$\frac{d^2y}{dx^2} = \frac{(g'(t) \cdot f''(t) - f'(t) \cdot g''(t))}{(g'(t))^2} \cdot \frac{1}{g'(t)}$$

$$\text{Put } t = 1$$

$$\frac{d^2y}{dx^2} = \frac{2}{1} = 2$$

$$\text{(D)} \quad L = \lim_{x \rightarrow 0^+} (-\ln x)^{\frac{1}{\ln x}}$$

$$e^{\lim_{x \rightarrow 0^+} \frac{\ln(-\ln x)}{\ln x}}$$

Applying L'Hospital rule

$$L = e^{\lim_{x \rightarrow 0^+} \frac{\frac{1}{-\ln x} \cdot \frac{-1}{x}}{\frac{1}{x}}}$$

$$= e^0$$

$$= 1$$

$$\Rightarrow \ln L = 0$$